

A Beginner-Friendly Modding Guide

Home Among the Stars

RimWorld Modlist & Learning Guide

Targeting RimWorld 1.6.4633 + All DLC

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1 Chapter 1: What is RimWorld

1.1 Learning Objectives

- Understand RimWorld's core game loop and how a colony grows from crash-landing to escape
- Choose the right storyteller, difficulty, and starting scenario for a first playthrough
- Evaluate colonists and pick a landing spot with good strategic properties
- Understand what each DLC adds and decide which to enable on a first run

1.2 The Core Game Loop

RimWorld is a colony survival simulator set on a distant planet. You start with a handful of survivors — called *pawns* — who have crash-landed with nothing but the clothes on their backs and a few scattered supplies. From there, the loop is simple:

Crash land → build shelter → gather food → survive threats → research better tech → expand your base → defend against bigger threats → build a spaceship and escape (or don't).

Each of your pawns has a unique backstory, personality traits, skills, and health conditions. They get hungry, tired, and moody. They fall in love, start fights, throw parties, and have mental breakdowns. Your job isn't to control them directly — you issue *orders* (build this wall, harvest those crops, attack that raider) and the pawns execute those orders based on their own priorities and personality.

RimWorld is **a story generator first, a strategy game second**. The AI storyteller adapts events to your colony's situation. Things will go wrong. Pawns will die in stupid, tragic, or heroic ways. That's not failure — it's the story. The game's loading screen even says it: *"RimWorld is a story generator, not a skill test. Losing is fun."*

Every colony is a unique narrative. Some colonies become peaceful farming communes. Others become cybernetic supersoldier factories. Some burn to the ground because a pyromaniac had a bad day. All of these are valid outcomes.

1.3 Choosing a Start Scenario

When you create a new game, you pick a starting scenario. Each gives you a different number of pawns and starting resources.

Scenario	Pawns	Resources	Difficulty	Recommendation
Crashlanded	3	Moderate (weapons, some steel, packaged meals, medicine)	Standard	Pick this for your first game
Rich Explorer	1	High (advanced weapon, gold, components)	Harder than it looks — one pawn doing everything	Try after your first colony
Naked Brutality	1	Nothing — literally naked, zero items	Very hard — one disease or bad fight ends it	Save for when you want a challenge

Crashlanded is the intended new-player experience. Three pawns let you cover all the essential skills (at least one good cook, one good builder, one good shooter) without being overwhelmed. Rich Explorer gives you one well-equipped pawn with good tech, but a single injury or mental break can spiral fast. Naked Brutality is RimWorld hard mode — you have nothing, you are nothing, and you will probably die to a squirrel.

1.4 Choosing a Storyteller

The storyteller is the AI that decides what happens in your colony — when raiders attack, when traders visit, when diseases strike, and when nothing happens at all. RimWorld has three core storytellers, each with a distinct personality.

Storyteller	Style	What to Expect
Cassandra Classic	Steady, predictable difficulty ramp	Events come at a consistent pace. As your colony grows wealthier, threats get bigger. She gives you time to recover between attacks.
Phoebe Chillax	Long peace, then sudden chaos	You'll go seasons with nothing happening — then a massive raid, a plague, and a solar flare all hit at once. The long quiet periods make the sudden disasters feel worse because you're not prepared.
Randy Random	Completely unpredictable	Anything can happen at any time, in any combination. Two raids back-to-back, followed by a cargo pod full of wooden feet, followed by nothing for a year. Entertaining, but not good for learning.

Recommendation: Cassandra Classic for your first colony. Her steady pacing teaches you the natural rhythm of RimWorld. You'll always be slightly behind where you need to be — that's by design — but you'll never feel blindsided. Save Randy for your third or fourth colony when you want chaos.

Choose **Strive to Survive** difficulty with Cassandra. It is labelled as the “intended RimWorld experience” and hits the sweet spot where you need to think about your decisions without feeling unfair.

1.5 Difficulty Settings

RimWorld offers a difficulty slider from **Peaceful** (no major threats) to **Losing is Fun** (the game actively tries to break you). Here's what each tier means in practice:

Difficulty	Threat Level	Who It's For
Peaceful	No raids or major hostiles	Tutorial only — you'll learn base-building but not survival
Community Builder	Light threats, generous mood bonuses	Relaxed playthroughs, storytelling without stress
Adventure Story	Moderate threats, small mood bonuses	A gentler introduction to combat
Strive to Survive	Serious threats, no bonuses	The intended experience — pick this
Blood and Dust	Heavy threats, mood penalties	For experienced players who want a fight
Losing is Fun	Maximum threats, harsh penalties	Self-explanatory — you will lose

You can also set a **custom difficulty** to tweak individual sliders — threat scale, colonist mood, crop yield, resource yield, and more. If Strive to Survive feels too punishing, drop it to Adventure Story rather than rage-quitting. You can change difficulty at any time from the options menu mid-game.

For a first colony, start on **Adventure Story** or **Strive to Survive** with Cassandra Classic. Adventure Story gives you more breathing room to learn; Strive to Survive teaches you good habits because mistakes have real consequences.

1.6 Colonist Selection Guide

Before you land, the game presents you with a pool of random colonists to choose from. You'll also see the option to **randomize** each slot — click it as many times as you like. Don't settle for terrible pawns just because they showed up first. A good starting team is the difference between a smooth first year and watching everyone starve by day 5.

1.6.1 Skills

Every pawn has 12 skills rated from 0-20, with higher being better. For your first colony, prioritize these skills in roughly this order:

1. **Plants** — Food is life. Someone needs to grow crops fast enough to feed everyone. Without a decent grower, you starve.
2. **Construction** — Walls, beds, workbenches, doors. If you can't build, you have no base.
3. **Cooking** — Raw food gives a mood penalty. Meals give mood bonuses. A good cook also reduces food poisoning risk.
4. **Shooting** — Raiders *will* come. Someone needs to hold a gun and hit what they aim at.
5. **Mining** — Steel and components come from mining. Without them, you can't build anything beyond wood.
6. **Medical** — Injuries happen. Infections kill untreated pawns. You need at least one pawn who can patch people up without making things worse.
7. **Social** — Trading prices, prisoner recruitment, and faction relations all depend on social skill.
8. **Intellectual** — Research unlocks electricity, better weapons, advanced medicine, and eventually the spaceship.
9. **Animals** — Taming and handling animals for hunting, hauling, and combat. Useful mid-game, not urgent early.
10. **Artistic** — Sculptures for beauty and mood. Important later, not early.
11. **Melee** — Useful for blocking doorways and hunting small game. Risky against armed raiders without good armor.
12. **Crafting** — Making weapons, armor, and components. Critical mid-game, not early.

You don't need every skill covered — **4-5 good colonists can cover the basics**. Look at your three Crashlanded pawns as a team: do they collectively have Plants, Construction, Cooking, Shooting, and Medical? If yes, you're in good shape.

1.6.2 Passions

Next to each skill in the colonist screen, you'll see zero, one, or two small flame icons. These represent **passion** — how much a pawn enjoys doing that work:

- **No flame:** The pawn does the task but gains XP very slowly.
- **One flame (interested):** Gains XP at 100% speed. Solid.
- **Two flames (burning passion):** Gains XP at 150% speed and gets a mood bonus while doing it. This is what you want.

A pawn with burning passion in Cooking will level up fast and stay happy while cooking. A pawn with no passion in Cooking will level slowly and might hate doing it. Prioritise passions over raw

skill numbers — a level 3 grower with burning passion will surpass a level 6 grower with no passion within a season.

1.6.3 Traits to Look For

Every pawn has 1-3 personality traits. Some are great, some are dealbreakers.

Good traits: - **Industrious** (+20% work speed) — they simply get more done, every single day. - **Fast Learner** (+50% XP gain) — they level up fast in everything. - **Jogger** (+20% move speed) — reaches fires, wounded pawns, and battle positions faster. Movement speed saves lives. - **Tough** (takes half damage) — the single best combat trait in the game. A tough pawn can tank hits that would kill anyone else. - **Iron-willed** (mental break threshold is very high) — they stay calm when everything is on fire. - **Sanguine** (permanent +12 mood bonus) — happy pawns don't have mental breaks. - **Kind** (won't insult others, gets mood from being kind) — reduces social fights, improves colony mood.

Dealbreaker traits — do NOT take these pawns on your first colony: - **Pyromaniac** — Will randomly start fires in your base during mental breaks. A wooden base + pyromaniac = colony-ending disaster. - **Gourmand** — Eats twice as much food and binges during food shortages. In a first colony that's already struggling with food, this is a death sentence. - **Wimp** — Incapacitated by minor pain. Falls over from a bruise. Useless in any fight and can't rescue others. - **Chemical Interest / Chemical Fascination** — Will randomly take drugs, get addicted, and go on binges. Addiction management is not a first-colony problem. - **Depressive / Pessimist / Volatile** — Permanent or frequent mood penalties. Mental breaks cascade into disasters for new players.

1.6.4 Age and Health

Avoid pawns over 60 unless their skills are spectacular. Old pawns develop age-related conditions — bad back (slower movement), cataracts (worse shooting), dementia (random confusion), and frail torso (vulnerable to one-shot kills). They can also develop these conditions mid-game, which you won't be equipped to handle early.

Check the **Health** tab on each pawn before finalising. Red flags: addiction (any kind), permanent injuries, carcinoma (cancer), or a missing limb. These can all be fixed eventually with bionic replacements, but not in the first year.

For your first colony, aim for **three pawns aged 20-50, with collective coverage of Plants/Construction/Cooking/Shooting/Medical, no dealbreaker traits, and at least some passion icons in their key skills.**

1.7 Biome Comparison Guide

The biome determines the climate, available plants/wildlife, diseases, and challenges of your map. Some biomes are forgiving. Others will kill you before you finish building your first room. Here is how they compare:

Biome	Difficulty	Growing Season	Wood	Unique Challenge
Temperate Forest	Easy	50-60 days/year	Abundant	None — best for first game
Boreal Forest	Medium	30-40 days/year	Moderate	Cold winters, short growing season
Arid Shrubland	Medium	40-50 days/year	Scarce	Heat waves, sparse wood, fewer animals

Biome	Difficulty	Growing Season	Wood	Unique Challenge
Tropical Rainforest	Medium	Year-round 60/60	Abundant	Constant diseases (malaria, sleeping sickness), aggressive predators
Desert	Hard	20-30 days/year	Very scarce	Extreme heat, almost no wood, few animals
Tundra	Hard	10-20 days/year	Scarce	Extreme cold, short growing, few animals
Extreme Desert	Very Hard	0-10 days/year	None	Heat, no wood, no soil, almost nothing lives here
Ice Sheet	Extreme	0 days/year	None	Perpetual -40°C or worse. No plants. No animals. Cannibalism or death.
Sea Ice	Impossible	0 days/year	None	There is nothing. Literally nothing. You will die.

Recommendation: Temperate Forest, on a tile with year-round growing (look for the “Growing Period: Year-round” label on the world map). This gives you the most forgiving setup: abundant wood, plenty of animals to hunt, mild temperatures, and a long growing season so your crops won’t die to an early frost.

If you want slightly more challenge without being punishing, Boreal Forest is a good second choice — you’ll need to stockpile food for winter and build heaters, but it’s manageable.

1.8 Choosing a Landing Tile

The world map is divided into hexagonal tiles. When you select a starting tile and click it, you’ll see detailed information about what’s there. Here are the features that matter most on your first colony:

- **Geothermal vents:** These are geysers of steam that let you build Geothermal Generators later in the game. Each one produces 3600W of free, infinite power — no fuel, no sun, no batteries needed. Look for tiles with **at least 2 vents**. Mark them during base planning — you’ll want to build your power grid near them.
- **Fertile soil:** Darker patches of ground where crops grow at 140% speed. This is a massive advantage for your food supply. A 5×5 field on fertile soil produces nearly as much as a 7×7 field on normal soil. You want at least some fertile soil inside your planned base area.
- **Road access:** Tiles connected to roads make caravan travel faster. If you’re near a road, traders visit more often and your own caravans move quicker. Ancient asphalt roads are the best; dirt roads are still helpful.
- **Mountains:** Building into a mountain gives you natural walls that raiders can’t break through. The downside: mountain bases risk **infestations** (giant insects bursting out of your walls) in the mid-to-late game. A tile with *some* mountain but plenty of open space gives you the best of both worlds.
- **Rivers:** A natural defensive barrier. Raiders move slower crossing water, giving your shooters more time to pick them off. Rivers can also power Watermill Generators for early-game electricity.
- **Proximity to settlements:** Neutral or friendly faction settlements within a few tiles mean nearby trading partners. You can send caravans to buy and sell goods without traveling across the entire world.

Recommendation: Find a Temperate Forest tile with year-round growing, at least **2 geothermal vents**, some **fertile soil**, and near a **road**. Don't overthink it — you can launch the ship from almost any tile. The important thing is to start playing.

1.9 Early Game (Days 1-15)

Your first two weeks set the trajectory for your entire colony. Here's the priority order:

Hour 1 — Shelter and sleeping spots. Pause the game immediately (Space bar). Find a defensible area near fertile soil and your geothermal vents. Build a single room — wood walls, wood door, wood roof — large enough for three sleeping spots. Use the Architect menu to designate beds or sleeping spots. A tiny wood shack is ugly but it keeps pawns from sleeping outside (which gives a massive mood penalty). Remove the “No Roof” zone from the area so it auto-builds a roof. In the meantime, your three survivors are standing around outside — get that room built first.

Day 1-2 — Food starts growing. Designate a growing zone (Architect → Zone → Growing Zone) on fertile soil. Plant **rice** — it is the fastest-growing crop (roughly 3 days to harvest) and will get food on the table before your starting packaged meals run out. Rice gives less yield per harvest than corn or potatoes, but it grows fast enough to prevent starvation while you get established. Plant potatoes on less fertile soil nearby. Set a stockpile zone for food and one for other items.

Day 2-5 — A real base emerges. Expand from the starter shack into separate rooms: one for sleeping, one for a kitchen/butcher table, one for a freezer (more on this below), and one for general storage. Build walls, floors, and doors. Build a horseshoe pin for recreation (pawns get miserable without it). Build a campfire or, if you've researched it, a wood-fired generator plus a cooler to create a walk-in freezer — this is what stops your food from rotting.

The freezer is the most important room in your base. Without it, meat rots in 2-3 days and meals rot in 3-4 days. A freezer is a room with double-thick walls, a cooler set to below 0°C, and an airlock (door → empty tile → door) to keep cold air in. Power it with a wood-fired generator or a couple of solar panels.

Day 3-7 — Research begins. Build a Research Bench. Set a pawn with good Intellectual skill to prioritise research. Your research order should be: **Batteries** (store power for night-time and eclipses) → **Solar Panels** (reliable daytime power) → **Stonecutting** (replace those wooden walls before they burn). You can also rush **Microelectronics** for the Comms Console and trade beacons, but that's a mid-game goal.

Day 5-10 — First raid. They're coming. Raiders scale based on your colony's wealth, so early raids are small — usually a single attacker with a knife or a poor-quality gun. Draft your pawns (select them and press R), position them behind walls or trees for cover, and shoot the raider before they reach melee range. If you have a pawn with decent Melee skill, hide them behind a door and ambush the raider as they walk through. After the fight, strip the raider's corpse for their weapon and any clothing — free gear.

Day 7-15 — Fireproofing and growing. Once you research Stonecutting, build a Stonecutter's Table and start making stone blocks. Replace your wooden walls with stone (any type — sandstone is fastest to cut, granite is strongest). A single fire in a wooden base spreads to every connected wooden wall and can destroy your entire base in minutes. Stone walls don't burn.

By day 15, you should have a small multi-room base, a functioning freezer, crops growing, basic research underway, and everyone armed with at least a poor-quality weapon. You should also have accepted your first refugee or rescue quest — an extra colonist is a huge boost this early.

1.9.1 Common Early-Game Mistakes

- **Building too big, too fast.** A massive base spreads your resources thin, takes forever to build, and is hard to defend. Start small, expand outward as you need rooms.

- **No freezer.** Meat spoils in 2 days. Meals spoil in 3. You will lose half your food to rot before you can eat it. Freezer first, always.
- **Wooden walls everywhere.** One lightning strike, one pyromaniac, one Zzztt event, and your base is gone. Stone walls = fire immune.
- **No recreation.** Pawns with no recreation get a “Recreation-starved” mood penalty that escalates quickly. Build a horseshoe pin on day one — it costs almost nothing.
- **Ignoring mood.** Look at the Mood tab on each pawn. If someone’s below 30%, they risk a mental break. Fix the problem — better food, nicer room, recreation, or more sleep.

1.10 Mid Game (Days 15-100)

By the mid game, your base should be self-sufficient: food is stable, you have a few extra colonists (aim for 6-10), and you’re starting to think about long-term survival rather than just getting through the next day.

Stable power grid. Solar panels + batteries are good but vulnerable to eclipses (no sun for days = dead batteries). Add a Geothermal Generator to each vent on your map. One geothermal vent produces 3600W forever — enough to run your entire base with no fuel and no downtime. Connect everything with power conduits. Add a few backup wood-fired generators for emergencies.

Defences. Raiders are getting bigger. A single wall and a couple of shooters won’t cut it any more. Build a **killbox** — a funnel-shaped entrance to your base that forces raiders into a narrow corridor where your shooters can concentrate fire on them from behind cover. Position colonists behind sandbags or barricades at the end of the funnel. Add traps — steel deadfall traps in a checkerboard pattern will cripple or kill raiders before they reach your firing line. Build walls around your base perimeter so raiders can’t walk in from any direction.

Prisoner recruitment. Capture downed raiders instead of killing them. Build a prison room (separate room with a bed, set it as a Prisoner Bed). Wardens (pawns with high Social) can attempt to recruit prisoners over time. A recruited raider joins your colony and brings their skills with them — this is your primary source of new colonists. Build individual cells rather than barracks: prisoners in barracks will constantly fight each other or attempt break-outs, while a private cell keeps them calm (and makes it easier to convert them if using Ideology).

Research priorities. After the survival basics, push toward: **Microelectronics** (Comms Console lets you call trade ships, Orbital Trade Beacon lets you sell items directly from your base) → **Fabrication** (craft components and advanced weapons yourself instead of relying on traders) → **Geothermal Power** (already mentioned).

Trading. Build an Orbital Trade Beacon over your main stockpile. This lets passing trade ships beam goods up in exchange for silver. If you’re near friendly settlements, send a caravan (two pawns, some pack animals, and your surplus goods) to trade in person. Sell excess food, crafted clothing, sculptures, or captured weapons. Buy components, advanced medicine, and psychic lances.

Key milestones: - Surviving a **siege** (raiders set up mortars outside your base and shell you from a distance). Counter with your own mortars, or sally out with your shooters to destroy the sieging party. - Building your first **multi-analyzer** to unlock advanced research. - Outfitting all combat colonists in flak vests and helmets — this is the baseline protection everyone needs. - Creating a dedicated hospital room with hospital beds, a vital monitor, and a stockpile of medicine.

1.11 Late Game (Days 100+)

Your colony is now established, wealthy, and well-defended. You have 10-15 colonists, advanced technology, and the resources to pursue one of the game’s win conditions — or simply keep building forever.

1.11.1 Win Conditions

RimWorld offers several ways to “beat” the game, but none are required. Many players build colonies that run for decades with no intention of ever leaving.

Ship to the Stars (base game). Research the entire shipbuilding technology tree, construct a spaceship from advanced components, and defend it through a 15-day reactor startup sequence. During those 15 days, the AI storyteller throws everything it has at you — escalating raids, mechanoid assaults, and sieges. If your ship survives and you put all your colonists aboard, you win. This is the classic RimWorld ending.

Royal Ascent (Royalty DLC). Gain favour with the shattered Empire faction by completing quests and hosting nobles. Raise a colonist to the rank of Count or Countess, then call a shuttle from the Empire to take you off-world. Along the way, your noble colonist gains psycasts — psychic powers that let you stun enemies, teleport allies, or call down lightning — but also demands increasingly luxurious bedroom and throne room requirements.

Archonexus (Ideology DLC). Sell your colony to an allied faction and restart with a portion of your wealth, research, and a few chosen colonists. Do this three times, each restart harder than the last, and you reach the Archonexus — a transcendent AI entity. This is the narrative deep-end, best attempted after you’ve played through the other endings.

No ending. Many players — perhaps most — never “win.” They build a thriving colony and enjoy the ongoing story of raids, weddings, disasters, and recovery. When the colony feels complete, they start a new one with different mods, a different biome, and a different story. This is a completely valid way to play.

1.11.2 Late-Game Milestones

- Defeating a **mechanoid cluster** (a self-contained base of killer robots that lands on your map with turrets, assemblers, and a shield).
- Surviving a **manhunter pack** (50+ enraged animals swarm your base — you either kill them all or hide indoors for days).
- Equipping front-line colonists with marine or recon armour.
- Installing bionic limbs and organs on injured pawns — lost an arm to a scyther? Replace it with a bionic arm that’s actually better than the original.
- Building a second colony on another tile — only recommended once your main base is fully automated and self-sufficient.

1.12 DLC Breakdown

RimWorld has five expansion packs. Each adds layers of complexity, and you can enable or disable them per save file. Here’s what each one does and when you should consider adding it:

Royalty. Adds the Empire faction (a shattered interstellar monarchy), psycasts (magic-like psychic powers), nobility titles, and a quest system with honour rewards. Royalty integrates cleanly with the base game — it doesn’t fundamentally change how you play, but adds new tools and a new faction.

When to add: After your first playthrough, or enable it from the start if you want faction variety and psycasts.

Ideology. Adds belief systems, rituals, roles (leader, moral guide), and memes (core beliefs). Every colonist follows an ideology that dictates what they find acceptable — some ideologies demand cannibalism, blind reverence for darkness, or tree worship. This completely reshapes every decision you make. Huge depth, significantly more complex. **When to add:** After you’re comfortable with the base game. Ideology is excellent but overwhelming if you’re still learning how freezers work.

Biotech. Adds children (colonists can have babies), mechanitors (controlling your own mechanoids), and xenogenes (genetic engineering — create custom humanoid races). This is the most integrated

and beloved DLC. The mechanics slot naturally into the base game and enrich every stage of play. **When to add:** Fine to enable on your first run. Biotech's additions appear organically and you can engage with them when you're ready.

Anomaly. Horror-themed expansion. Adds void monsters, monolith events, flesh beasts, and reality-warping threats. Your colonists can be captured, mutated, or driven insane by entities from beyond. Much more dangerous than the base game. **When to add:** Wait until you've defended against regular raids, mech clusters, and sieges. Anomaly's threats are designed to challenge experienced players.

Odyssey. A narrative-focused expansion with new factions, quest chains, and territory mechanics. Adds fishing and ocean-based content. **When to add:** Fine to enable from the start — it enriches world generation and adds variety without dramatically increasing difficulty.

Recommendation for your first run: Enable **Biotech** and **Odyssey**. Add **Royalty** if you want psycasts. Save **Ideology** and **Anomaly** for a second or third playthrough when you're ready for more complexity.

1.13 What is Modding?

Modding means modifying the game with community-created content. RimWorld has one of the most active modding communities in gaming — tens of thousands of mods add everything from bugfixes to entire new game systems.

Here's what you need to know before you start:

- **Mods come from the Steam Workshop.** You subscribe to them on Steam, they download automatically, and RimPy (from Chapter 2) handles activating and sorting them. No manual file downloads, no zip extraction.
- **Load order matters.** Mods load from top to bottom when the game starts. If two mods change the same thing (say, how crafting menus look), the lower one wins. This is called "load order" and getting it wrong causes bugs or crashes. RimPy auto-sorts your load order correctly — you don't need to understand the details.
- **Frameworks come first.** Some mods are "libraries" — they provide tools that other mods use, but don't change the game themselves. The two essential ones are **Harmony** (a code-patching library) and **HugsLib** (a shared toolkit for mod authors). These must load before any mod that depends on them. RimPy handles this automatically.
- **You can create mod profiles.** RimPy lets you save different mod sets as "profiles." You can have a profile for this guide's modlist, another for a vanilla-like experience, and another for a challenge run. Switching profiles takes one click.
- **Some mods can't be removed mid-game.** Check each mod's Workshop page for compatibility notes. If it says "NOT safe to remove mid-save," removing it will corrupt your save file. When in doubt, start a new colony if you want to change your mod set.
- **RimWorld mods mostly play well together.** Unlike games like Skyrim where mod conflicts are constant, RimWorld's architecture makes mod conflicts relatively rare. The curated list in this guide has been verified for compatibility.

1.14 Next Steps

Proceed to [Chapter 2: Setup](#).

2 Chapter 2: Setup

2.1 Learning Objectives

- Install RimWorld and verify all DLC are active
- Install and configure RimPy Mod Manager
- Navigate the Steam Workshop to find and subscribe to mods
- Understand load order and how RimPy auto-sort works
- Set up mod profiles for different playthroughs

2.2 Installing RimWorld

RimWorld is available exclusively through Steam on Windows. If you haven't installed it yet:

1. Open Steam and search for "RimWorld" in the store.
2. Purchase the base game.
3. Install it through Steam (default location: C:\Program Files (x86)\Steam\steamapps\common\RimWorld).
4. Purchase and install the DLCs — Royalty, Ideology, Biotech, Anomaly, and Odyssey. You can install them all at once; you'll choose which to activate per save file.

Before doing anything with mods, launch RimWorld **once in vanilla** (no mods) to verify everything works. From the main menu: 1. Click **New Colony**. 2. The scenario selector will appear — you'll see all your owned DLCs listed as available. 3. Start the tutorial scenario if you've never played before. Or just create any colony to verify the game runs smoothly. 4. Exit the game.

This first launch creates RimWorld's configuration folders, which is what RimPy needs to detect your installation. If you skip this step, RimPy won't be able to find the game.

2.3 Installing RimPy

RimPy Mod Manager is a community-built tool that replaces RimWorld's built-in mod manager. It does three things the built-in manager cannot: 1. **Auto-sort** your mod load order correctly with one click. 2. **Warn you** about missing dependencies and known conflicts before you launch the game. 3. **Download mods** from Steam Workshop without needing to launch the game first.

2.3.1 Installation Steps

1. **Download RimPy.** Go to the [RimPy releases page](#). Download the latest release — look for the .exe installer or the portable .zip file for Windows. As of this writing, the latest version is RimPy Mod Manager 1.1.x.
2. **Run the installer.** If you downloaded the .exe, double-click to install. If you downloaded the portable .zip, extract it to a folder like C:\RimPy or your Documents folder. RimPy works from any location.
3. **First launch.** Open RimPy. It may take a moment to scan your Steam library. If it can't find RimWorld automatically, point it to your RimWorld install folder manually through the Settings menu.
4. **Steam integration.** RimPy needs to communicate with Steam to download mods. If you're logged into Steam on the same machine, this should work automatically. If RimPy shows a Steam connection error, restart Steam and try again.

2.3.2 What You'll See

When RimPy opens, you'll see: - A list of your subscribed mods on the left. - Your active mod list (the "load order") on the right. - Buttons at the top for sorting, saving, and managing profiles. - Colour-coded dependency warnings if anything is missing.

The interface looks busy on first glance, but you'll mostly use three buttons: **Sort**, **Save**, and **Run**. Everything else is information.

2.4 Navigating the Steam Workshop

All RimWorld mods are hosted on the [Steam Workshop](#). Here's how to find and subscribe to the ones in this guide:

1. **Open a mod's Workshop page.** Each mod entry in the following chapters includes a clickable name that links to its Workshop URL. The URLs follow the pattern `https://steamcommunity.com/sharedfiles/filedetails/?id=<WORKSHOP_ID>`.
2. **Read the description.** Before subscribing, scroll through the mod's description. Look for:
 - **Required items** — other mods or DLC that must be installed for this mod to work. A mod might say "Requires: Royalty DLC" or "Requires: HugsLib." RimPy also warns you about missing dependencies, but it's good to know before subscribing.
 - **Known incompatibilities** — mods that this one conflicts with. The curated list in this guide avoids known conflicts, but always check if you're adding mods outside the list.
 - **Version compatibility** — every mod in this guide has been verified for RimWorld 1.6. If a mod's Workshop page doesn't mention 1.6, skip it.
3. **Click Subscribe.** The green "Subscribe" button on the Workshop page adds the mod to your Steam library. Steam downloads it automatically in the background. You can see download progress in the Steam client's Downloads page.
4. **Wait for the download.** Some mods are a few kilobytes; others are hundreds of megabytes (especially graphics and sound mods). Give large mods a minute to finish downloading before launching RimPy.
5. **Refresh RimPy.** After subscribing, click the refresh button in RimPy (or restart it) and the new mod appears in your library.

2.4.1 Collections

Some mod authors create **Workshop Collections** — curated lists of mods that work together. If you see a collection linked from a mod's page, it's worth browsing, but don't blindly subscribe to everything in it. The modlist in this guide is more carefully curated for a beginner's first modded run.

2.4.2 Scanning for Updates

Steam keeps mods up to date automatically. However, some mods receive updates that change how they work or add new requirements. RimPy will flag mods whose Workshop version doesn't match your local version. If an update breaks your save, RimPy can roll back mods to earlier versions using its backup system.

2.5 Understanding Load Order

When RimWorld starts, it loads mods in a specific sequence from top to bottom. This sequence — the **load order** — is the single most important thing to get right. An incorrect load order causes crashes, broken features, and corrupted saves.

Here's how it works:

- **Mods that define the rules load first.** Harmony (code patching) and HugsLib (shared library) must always be at the very top of your load order. Every mod that uses them expects them to be loaded already. If Harmony isn't first, half your modlist won't work.
- **Frameworks and libraries load next.** Mods like JecsTools, ResearchPal, and Eccentric Tech - Core provide tools that other mods use. They must load before the mods that depend on them.
- **Content mods load last.** Anything that adds items, factions, animals, weapons, or quests. These use the frameworks from above to add their content.
- **Later mods override earlier ones.** If Mod A adds a crafting recipe for a steel sword, and Mod B also adds a recipe for a steel sword, the one from Mod B wins because it loads later. This is called

“patching” and is the primary way mods interact — one mod tweaks another’s behaviour by loading after it.

This sounds complicated, and it is. **That is why RimPy exists.** RimPy reads every mod’s metadata (its dependencies, what it patches, what it conflicts with) and calculates the correct load order automatically. You click one button. It sorts hundreds of mods in seconds. You never need to manually drag mods into position.

If you ever see advice about “just drag Harmony to the top,” ignore it — that’s advice for people not using RimPy. With RimPy, the auto-sort handles everything, including edge cases you’d never think of on your own.

2.6 Setting Up RimPy

Now that you understand what RimPy does, let’s configure it properly.

2.6.1 Step 1: Verify Your Mod List

Open RimPy. The left panel shows all mods you’ve subscribed to. The right panel shows your active load order. For a fresh install:

1. Click **Tools** → **Refresh Mods** (or the refresh icon) to ensure RimPy has the latest list.
2. In the left panel, verify that Harmony and HugsLib appear (if you’ve subscribed to them yet — if not, they’re covered in Chapter 3).
3. Ignore the right panel for now — it’s empty because you haven’t activated anything yet.

2.6.2 Step 2: Create a Profile

Profiles let you switch between different mod sets without manually activating and deactivating mods. This is useful for having a separate profile for this guide’s modlist versus a vanilla profile for testing.

1. Click the **Profiles** dropdown near the top of RimPy.
2. Select **New Profile**.
3. Name it “Home Among the Stars” or any name you prefer.
4. Click **OK**.

You now have an isolated profile. All mod activations and sorting in this profile won’t affect any other profiles. You can create additional profiles later — for example, a “Vanilla” profile with no mods for when you want the pure RimWorld experience.

2.6.3 Step 3: The Sort Button

The **Sort** button (or Ctrl+S) is RimPy’s core feature. It scans every activated mod, determines the correct load order, and rearranges them:

1. Activate a few mods by double-clicking them in the left panel (they move to the right panel). Or activate all mods in your current profile.
2. Click **Sort**.
3. RimPy displays a green checkmark if the sort succeeds, or a red warning with a description of what went wrong (missing dependency, version conflict, etc.).

Read the warnings. RimPy’s error messages are specific and actionable — it will tell you exactly which mod is missing a dependency and what that dependency is.

2.6.4 Step 4: Save and Run

After sorting: 1. Click **Save** (or Ctrl+S again) to save the current load order to your profile. 2. Click **Run** to launch RimWorld with this load order.

RimWorld will start up. You'll see the main menu with a "Mods" button. Click it to verify your mod list — it should match what you set up in RimPy. If a mod is missing, go back to RimPy, refresh, and check that you're subscribed.

2.6.5 RimPy Quick Reference

Action	How
Sort load order	Click Sort button (or Ctrl+S)
Save profile	Click Save
Launch game	Click Run
Refresh mod list	Click Refresh (circular arrow icon)
Activate/deactivate a mod	Double-click it
Search mods	Type in the search bar above the left panel
Check for updates	RimPy shows an orange update icon on outdated mods
Reorder manually	Drag mods in the right panel (but auto-sort is better)

2.7 Your First Mod Install

Let's walk through installing one mod from start to finish. We'll use **Harmony** as the example because nearly every mod in this guide depends on it.

2.7.1 Step 1: Subscribe on Steam Workshop

1. Open the [Harmony Workshop page](#) in your browser.
2. Click the green **Subscribe** button.
3. Steam will download Harmony in the background. The download is tiny (under 1 MB) — it should complete almost immediately.
4. Verify in the Steam client's **Downloads** page that the download finished.

2.7.2 Step 2: Activate in RimPy

1. Open (or switch to) RimPy.
2. Click **Refresh** if Harmony doesn't appear in your mod list.
3. Double-click **Harmony** in the left panel. It moves to the right panel — it's now active on your current profile.
4. Click **Sort**. Since it's the only mod active, the sort is trivial.
5. Click **Save**.

2.7.3 Step 3: Launch and Verify

1. Click **Run** in RimPy to launch RimWorld.
2. At the main menu, click the **Mods** button. You should see Harmony listed and active.
3. If it's marked with a green checkmark or no error indicator, it's loaded correctly.
4. Start a new colony and look at the debug log (Ctrl+` to open the console). You should see Harmony's startup message confirming it initialised. If there are any red error lines, note them and check the troubleshooting section in Chapter 15.
5. Exit RimWorld.

That's the complete install flow for one mod. Every mod in subsequent chapters follows the same pattern: **Subscribe** → **Refresh in RimPy** → **Activate** → **Sort** → **Save** → **Run**. You'll get fast at it.

Repeat this process for HugsLib (Workshop ID 818773962) as a second exercise. Harmony + HugsLib are the “always installed” pair — they form the foundation everything else builds on.

2.7.4 Before Moving On

At this point you should have: - RimWorld installed and launched once in vanilla mode. - RimPy Mod Manager installed and able to see your mods and launch RimWorld. - Harmony installed and verified as working. - A basic understanding of the Workshop → RimPy → Sort → Launch pipeline.

Don't worry about subscribing to every mod yet — each chapter will introduce mods as you read, and you'll install them one group at a time. The goal is to build your modlist progressively, understanding what each mod adds before installing the next one.

2.8 Next Steps

Proceed to [Chapter 3: Framework](#).

3 Chapter 3: Framework Mods

3.1 Learning Objectives

- Install the core libraries that almost every mod depends on
- Understand what each framework does
- Verify they load correctly

3.2 Overview

Framework mods don't add content themselves — they provide APIs that other mods depend on. They must load first in your mod order (RimPy handles this automatically).

3.3 The Mods

3.3.1 Harmony

- **What it does:** Harmony is a code-patching library that lets mods modify RimWorld's built-in behaviour without overwriting game files. It's the foundation nearly every RimWorld mod depends on.
- **Why it's here:** Without Harmony, mods can't safely change how the game works. It enables other mods to add features, fix bugs, and improve UI without touching RimWorld's original code. It's the single most important mod in any modded setup.
- **Not OP because:** Pure framework — zero gameplay impact. It doesn't add, remove, or change anything in the game itself. It simply makes other mods possible.
- **Install:** OK to install now. Must be at the very top of your load order (RimPy handles this). Safe to add mid-save.

3.3.2 HugsLib

- **What it does:** HugsLib is a shared library mod that provides common utilities for other mods — settings menus, logging tools, and helper functions. Also includes the Log Publisher (Ctrl+F12 to upload logs for troubleshooting).
- **Why it's here:** Many mods in this guide (Allow Tool, Defensive Positions, and others) depend on HugsLib. Installing it once centrally prevents each mod from shipping its own copy. Best practice: load it right after Harmony.
- **Not OP because:** Pure library — no gameplay content. It provides infrastructure for other mods but doesn't change anything itself.
- **Install:** OK to install now. Place right after Harmony in load order. Safe to add mid-save.

3.3.3 XML Extensions

- **What it does:** Provides XML patching utilities that let other mods modify game data cleanly without overwriting files. Enables mods to add new texture color variants, floor types, and other data-driven content.
- **Why it's here:** Several graphical and content mods (including CCP's Stone Tweaks Vanilla) now require XML Extensions as of RimWorld 1.6. Its patching system is the modern replacement for older XML-based framework approaches.
- **Not OP because:** It's a pure framework — no gameplay content, no balance impact. It simply enables other mods to work.
- **Install:** OK to install now. Safe to add mid-save.

3.3.4 Vanilla Expanded Framework

- **What it does:** A core modding library required by the entire Vanilla Expanded series. Provides shared code for custom gizmos, UI systems, hediff mechanics, faction management, base generation, and more. Also removes the hard cap of 12 factions set by the base game.
- **Why it's here:** Many mods in this guide (Vanilla Furniture Expanded, Vanilla Cooking Expanded, Vanilla Weapons Expanded, and more) depend on Vanilla Expanded Framework as their backbone.

It centralises shared code so each mod doesn't have to ship its own copy, keeping the modlist lean and compatible.

- **Not OP because:** Pure framework — no gameplay content, no items, no mechanics. It provides infrastructure for other mods to build on. The faction-cap removal is the only user-facing change, and it only matters if other mods add factions.
- **Install:** OK to install now. Place after XML Extensions and before Eccentric Tech - Core in load order. Safe to add mid-save.

3.3.5 Eccentric Tech - Core

- **What it does:** Adds an advanced AI mainframe (Aurora Core), nanofabrication facilities, and Second Wave materials (Nanoweave, HD Hex Cells). Unlocks the entire Eccentric Tech research tree.
- **Why it's here:** Required dependency for Eccentric Tech - Advanced Shields. Also enables optional upgrades for Fusion Power and other Eccentric mods. The Aurora Core gives the research bench a personality — Aurora, a starship AI — which adds lore flavour without gameplay impact.
- **Not OP because:** The nanofabrication chain requires significant research investment and advanced resources. The Aurora Core must be built and connected to a Hi-Tech Research Bench, consuming space and components.
- **Install:** OK to install now. Safe to add mid-save.

3.4 Installation Checkpoint

 OK to install all mods in this chapter now. They are safe to add mid-save.

3.5 Next Steps

Proceed to [Chapter 4: Bugfixes & Performance](#).

4 Chapter 4: Bugfixes & Performance

4.1 Learning Objectives

- Fix known vanilla bugs that can ruin a colony
- Optimize game performance for smooth 4K gameplay
- Understand what impacts TPS (ticks per second)

4.2 Overview

4.3 The Mods

4.3.1 Forced Rearmament

- **What it does:** Ensures colonists and slaves automatically rearm themselves with their dropped weapon after recovering from a knockdown. Vanilla pawns sometimes forget their weapons — this mod makes them reliably pick it back up.
- **Why it's here:** It's a common and frustrating vanilla bug — a pawn gets knocked down, gets back up, and wanders off unarmed, leaving their weapon on the ground. You don't notice until the next raid. Forced Rearmament fixes this silently. Drafting a pawn or having them rescued cancels the rearm, so you keep manual control when it matters.
- **Not OP because:** Only affects rearming behaviour — no stat changes, no speed boosts, no combat advantages. The same weapon, same pawn, same DPS. Enemies are unaffected (they behave exactly as vanilla). It simply fixes a bug where pawns forget to pick up their own gear.
- **Install:** OK to install now. Safe to add/remove mid-save.

4.4 Installation Checkpoint

- ✓ OK to install all mods in this chapter now.

4.5 Next Steps

Proceed to [Chapter 5: UI / HUD](#).

5 Chapter 5: UI / HUD

5.1 Learning Objectives

- Improve RimWorld's interface for comfortable 4K use
- Reduce unnecessary clicks with smarter menus and tooltips
- Get better visibility into colony status

5.2 Overview

5.3 The Mods

5.3.1 Camera+

- **What it does:** Extends zoom range (both in and out), smooths camera movement and edge scrolling, adds far-zoom labels, marker dots/silhouettes, and keyboard shortcuts for saving/loading camera positions.
- **Why it's here:** Vanilla RimWorld's zoom range is restrictive — you can't see the whole map at once or zoom in close enough to appreciate pawn details. At 4K, being able to zoom out for battlefield overview and zoom in for inspection is essential. The dot/silhouette system also makes pawns visible at max zoom.
- **Not OP because:** Pure camera control. No gameplay changes, no balance impact. You see the same things, just from better distances.
- **Install:** OK to install now. Safe to add mid-save.

5.3.2 BetterInfoCard

- **What it does:** Expands the vanilla info card (the panel that opens when you click the "i" button) — makes it resizable, collapsible, and keep multiple info cards open at once. Adds stat comparison, stat favorites, and a hotkey toggle. Opening an info card no longer pauses the game.
- **Why it's here:** The vanilla info card is tiny, pauses the game unnecessarily, and only shows one at a time. If you want to compare two weapons or items, you have to memorise the first card's stats before opening the second. BetterInfoCard fixes all of this.
- **Not OP because:** It's purely a UI improvement — it surfaces information that's already in the game. No gameplay changes.
- **Install:** OK to install now. Safe to add mid-save.

5.3.3 Damage Indicators [1.6]

- **What it does:** Displays floating damage numbers above pawns, animals, and enemies when they take damage. Configurable — you can filter to only show damage from pawns (hiding natural decay/fire damage).
- **Why it's here:** Vanilla gives you no feedback on how much damage your attacks are doing. Floating numbers help you evaluate weapons, compare damage output, and understand combat outcomes without digging through the log. Especially useful for a new player learning which weapons are effective.
- **Not OP because:** Pure visual feedback — it doesn't change damage values, hit chances, or any combat mechanic. What you see is what was always happening.
- **Install:** OK to install now. Safe to add mid-save. Disable the "show all damage" setting in large raids if you notice lag.

5.3.4 Allow Tool

- **What it does:** Adds a suite of selection and order tools — Allow All, Haul Urgently, Finish Off, Force Harvest, Select Similar, and more. Accessed through new toolbar buttons and right-click menus.

- **Why it's here:** Vanilla RimWorld requires you to click each item individually to unforbid it after a battle, or manually mark each plant for harvest. Allow Tool turns multi-click chores into single-click operations, saving hundreds of clicks per session.
- **Not OP because:** It only automates *existing* actions you could already do manually — it doesn't make anything faster, cheaper, or stronger. The pawns still need to walk over and do the work.
- **Install:** OK to install now. Safe to add mid-save. Requires HugsLib (already in your load order from Chapter 3).

5.3.5 RimHUD

- **What it does:** Replaces the vanilla bottom-left pawn card with a comprehensive info panel showing stats, equipment, health, skills, mood, and needs — all on one screen. Fully customisable layout.
- **Why it's here:** The vanilla pawn card hides most useful information behind sub-tabs. RimHUD surfaces everything at once — you can see a pawn's shooting accuracy, move speed, equipped weapon DPS, health conditions, and current job without clicking anywhere. Invaluable at 4K where the tiny vanilla card is even harder to read.
- **Not OP because:** Pure information display. It shows stats that already exist in the game — nothing is added, buffed, or changed. You just don't have to hunt for the data anymore.
- **Install:** OK to install now. Safe to add mid-save.

5.3.6 Better Workbench Management

- **What it does:** Overhauls the workbench bill management interface — adds a searchable bill list, drag-and-drop reordering, bill copying, and a cleaner layout for managing multiple production queues.
- **Why it's here:** Once you have 5+ workbenches each with multiple bills, the vanilla interface becomes a scrolling nightmare. Better Workbench Management turns bill management into a fast, searchable, organised workflow.
- **Not OP because:** Pure UI improvement. It doesn't change production speed, material costs, or bill limits — just how you find and organise them.
- **Install:** OK to install now. Safe to add mid-save.

5.3.7 Designator Shapes

- **What it does:** Adds a shape palette to the planning/designation tools — draw lines, rectangles, ellipses, circles, pentagons, hexagons, grids, and pre-set shapes (sun lamp, trade beacon). Includes undo/redo and a flood-fill tool.
- **Why it's here:** Building curved or diagonal structures in vanilla requires placing each tile individually. Designator Shapes lets you plan and build in geometric patterns — essential for aesthetic bases, efficient strip-mining grids, and precise defensive layouts.
- **Not OP because:** Pure building tool. It doesn't add resources, speed construction, or change game mechanics. It just makes placing blueprints faster and more precise.
- **Install:** OK to install now. Safe to add mid-save.

5.3.8 Dynamic Trade Interface

- **What it does:** A high-performance, customisable trade window — resize, reorder columns, search/filter with RegEx, save filter presets, sort by any column, buy/sell max affordable with one click. Does not replace vanilla; both are accessible via a hotkey.
- **Why it's here:** The vanilla trade window is slow and cluttered, especially with modded items. Dynamic Trade Interface makes trading fast — search for what you need, see unit prices clearly, and process bulk trades in seconds instead of minutes.
- **Not OP because:** Pure UI improvement to an existing system. Trade prices, availability, and mechanics are unchanged. It just makes the interface faster.
- **Install:** OK to install now. Safe to add/remove mid-save.

5.3.9 Range Finder

- **What it does:** Select one or more pawns/turrets, press a hotkey, and their weapon range is displayed on the map as a coloured overlay. Press twice to keep it visible until toggled off.
- **Why it's here:** Vanilla doesn't show you weapon ranges — you estimate by eye. Range Finder lets you position firing lines, place turrets, and plan hunting grounds with precision. Essential for effective defensive setup.
- **Not OP because:** Pure visual overlay. Range values are the same as vanilla — they're just visible now.
- **Install:** OK to install now. Safe to add mid-save.

5.3.10 Power Tab 2 - 1.6 Bug Fixed Version

- **What it does:** Overhauls the power grid information window — shows connected consumers, battery charge, production vs. consumption, and power storage details in an organised, expandable UI.
- **Why it's here:** Vanilla's power tab shows a single number (net power). When you have chemfuel generators, solar panels, batteries, and turrets all on one grid, you need to see where power is going. Power Tab 2 gives you the full picture.
- **Not OP because:** Information-only. It doesn't change power production, consumption, or battery mechanics. It surfaces data the game already tracks internally.
- **Install:** OK to install now. Safe to add mid-save. This is a bug-fixed fork that works on 1.6.

5.3.11 DDI - Deep Drill Indicator

- **What it does:** Adds a visible overlay on all scanned deep drill resources once your ground-penetrating scanner is powered. No more hovering over each drill site to check what's there.
- **Why it's here:** After scanning, you need to remember or re-hover over every underground deposit to find the resource you need. DDI puts the resource type and count right on the map — a huge time-saver during mid-game resource scouting.
- **Not OP because:** The resources must still be scanned with a powered ground-penetrating scanner. It only displays what you've already discovered — no free information.
- **Install:** OK to install now. Requires a powered ground-penetrating scanner to function.

5.3.12 RandomPlus

- **What it does:** Lets you set minimum specifications for pawn generation — minimum skill levels, required traits, age range, and more. The randomise button keeps rerolling until it finds a pawn meeting your criteria (with a configurable max reroll limit).
- **Why it's here:** Vanilla pawn generation is pure RNG — you can click "randomise" for 20 minutes without seeing a pawn with both Plants and Construction. RandomPlus saves hours of clicking without letting you hand-pick super-pawns.
- **Not OP because:** You're still working within the game's normal pawn generation pool — you just set a floor. A pawn with 8+ in Plants and no pyromania is a good start, not a god roll. The max reroll limit prevents infinite searching.
- **Install:** OK to install now. Safe to add mid-save.

5.3.13 Day Counter

- **What it does:** Adds a persistent on-screen display showing how many days have passed in your colony. Fully configurable position and appearance in mod settings.
- **Why it's here:** The vanilla game tracks days internally but doesn't show them prominently. Knowing what day you're on helps plan around seasonal growing windows, raid cooldowns, and quest timers. Particularly useful for new players learning the game's pacing.
- **Not OP because:** Pure information display. The day counter already exists in the game (visible in the history tab) — this just puts it on screen.
- **Install:** OK to install now. Safe to add mid-save.

5.3.14 Fahrenheit and Celsius

- **What it does:** Displays both Fahrenheit and Celsius in the temperature readout. Which scale appears first depends on your game settings.
- **Why it's here:** RimWorld discussions, guides, and wikis use Celsius, but if you grew up with Fahrenheit, you have to mentally convert every temperature. This mod shows both at once — no more wondering what 20°C “feels like.”
- **Not OP because:** Pure display change. Temperature mechanics are untouched — it just shows you the same number in two units.
- **Install:** OK to install now. Safe to add mid-save.

5.3.15 Tainted Renaming

- **What it does:** Moves the “T” prefix from the start of tainted apparel names to the end — “Tainted Shirt” becomes “Shirt (Tainted)”. Configurable in settings.
- **Why it's here:** Alphabetical sorting puts all tainted items together at the top of your lists. Moving the tag to the end sorts items by their actual name, so shirts are with shirts and pants are with pants. A small change that makes inventory management noticeably cleaner.
- **Not OP because:** Purely cosmetic text change. No gameplay impact whatsoever.
- **Install:** OK to install now. Safe to add mid-save.

5.4 Installation Checkpoint

- ✓ OK to install all mods in this chapter now.

5.5 Next Steps

Proceed to [Chapter 6: Graphics](#).

6 Chapter 6: Graphics

6.1 Learning Objectives

- Improve visual clarity at 4K resolution
- Enhance RimWorld's aesthetics without changing gameplay
- Keep good performance while improving visuals

6.2 Overview

6.3 The Mods

6.3.1 AI Upscaled Textures - Core

- **What it does:** AI-upscaled textures for all Core, Royalty, Ideology, Biotech, and Anomaly assets — pawns, plants, items, buildings, overlays, and more. Everything gets a resolution bump.
- **Why it's here:** At 4K, RimWorld's vanilla textures can look pixelated and blurry. AI Upscaled Textures sharpens the entire game world — pawns, items, and buildings all gain crispness without changing the art style. The mod author recommends pairing it with Graphics Settings+ to reduce VRAM usage.
- **Not OP because:** Purely cosmetic texture replacement. No gameplay changes. The AI upscaling has limitations (it can't add detail that wasn't there) but it removes the "jagged edge" look at high resolution.
- **Install:** OK to install now. Safe to add mid-save. Recommends Graphics Settings+ mod for performance.

6.3.2 Vanilla Pawns Retextured

- **What it does:** Replaces human pawn textures with crisp 512×512 versions. Also covers Biotech gene-related textures. Designed as a base for alien race textures but works beautifully as a standalone visual upgrade.
- **Why it's here:** Pawns are the characters you watch for hundreds of hours — they deserve to look sharp. At 4K, vanilla pawn textures can appear soft. Vanilla Pawns Retextured gives them clean, defined edges without changing the art direction.
- **Not OP because:** Pure texture replacement. The mod changes nothing about pawn stats, behaviour, or gameplay. Its textures will override other texture packs loaded before it in the load order.
- **Install:** OK to install now. Safe to add mid-save. Load after AI Upscaled Textures if using both, so VPR's pawn textures take priority.

6.3.3 CCP's Stone Tweaks Vanilla

- **What it does:** Applies colour tweaks to all vanilla stone types (sandstone, granite, limestone, slate, marble) so they look more distinct and natural. Includes built-in patches for Vanilla Royalty, Vanilla Ideology, Expanded Floors, and more. Colours are fully customisable via the mod settings menu.
- **Why it's here:** Vanilla stone types all blend into a muddy grey-brown at 4K resolution. Distinct colours make stone floors, walls, and sculptures visually readable — you can immediately tell granite from marble at a glance.
- **Not OP because:** Purely cosmetic. Nothing changes balance, game mechanics, or resource values. The mod only touches textures and colours.
- **Install:** OK to install now. Safe to add mid-save. Requires XML Extensions (already in your load order from Chapter 3).

6.3.4 Floor-Mounted Aquarium

- **What it does:** Adds floor-mounted aquariums in 3 sizes that emit light, increase room beauty, and periodically produce fish. Powered buildings that function as both decoration and a passive food source. Works with either Vanilla Fishing Expanded or Odyssey DLC's fishing mechanics.
- **Why it's here:** Bases look sterile without decorative touches. Aquariums add visual warmth to rec rooms, dining halls, and hospitals. The passive fish production is a small bonus, not the main draw — they're a beauty-building joy item first.
- **Not OP because:** Aquariums require power, resources to build, and space. Fish production is slow and minor — it won't replace hunting or farming. They're purely cosmetic with a negligible food side-effect. The beauty boost only applies while powered.
- **Install:** OK to install now. Safe to add mid-save.

6.3.5 Animal Variety Coats

- **What it does:** Adds colour and pattern variation to vanilla animals — muffalo, wargs, timber wolves, rhinos, thumbos, and 20+ other species no longer look identical. Each animal spawns with randomised coat colours and patterns, making herds and packs visually distinct.
- **Why it's here:** When every muffalo in your herd looks identical, they blur into a single visual blob — especially at 4K where the repetition is more noticeable. Animal Variety Coats makes each animal feel individual. Spotting your favourite muffalo in a pack becomes possible. Pairs well with Vanilla Pawns Retextured and AI Upscaled Textures for a complete visual refresh.
- **Not OP because:** Purely cosmetic texture variation — no stat changes, no gameplay impact. Leather colour and wool colour are unchanged (leather tint is set per-species, not per-individual). Only affects newly spawned animals.
- **Install:** OK to install now. Safe to add mid-save — only newly spawned animals will show variation. Compatible with all creature mods from Chapter 12.

6.4 Installation Checkpoint

- ✓ OK to install all mods in this chapter now.

6.5 Next Steps

Proceed to [Chapter 7: Audio & Music](#).

7 Chapter 7: Audio & Music

7.1 Learning Objectives

- Expand RimWorld's soundtrack with new music
- Reduce irritating sound effects for a more comfortable experience
- Customise the auditory atmosphere of your colony

7.2 Overview

RimWorld's soundtrack is good, but hearing the same tracks across hundreds of hours of gameplay gets repetitive. These mods add new music and clean up small audio annoyances so you can enjoy the soundscape longer.

7.3 The Mods

7.3.1 Expedition 33 Music

- **What it does:** Adds 154 songs from the Clair Obscur: Expedition 33 soundtrack into RimWorld's music rotation. Integrates with the vanilla music system — tracks play during appropriate moments (combat, ambient, etc.).
- **Why it's here:** Vanilla RimWorld has a limited track count. After 50+ hours, you've heard every song dozens of times. Expedition 33's orchestral soundtrack fits RimWorld's sci-fi tone naturally and dramatically expands your musical palette.
- **Not OP because:** Pure audio — no gameplay, balance, or performance impact. Music plays in the background exactly like vanilla tracks. If you don't like a track, it won't play again.
- **Install:** OK to install now. Safe to add mid-save. For the best experience, the author recommends using Music Manager (Continued) by Zaljerem to control playback.

7.3.2 No Time Change Sounds

- **What it does:** Mutes the sound effect that plays when you change game speed (pause → slow → normal → fast). That's all it does.
- **Why it's here:** If you're like most RimWorld players, you change time settings constantly — pausing to issue orders, speeding through travel, slowing down for combat. The “click-chime” on every speed change adds up to hundreds of unnecessary sound triggers per session. This mod silences them.
- **Not OP because:** Muting a UI sound effect has zero gameplay impact. It's pure comfort — the game plays identically, just more quietly.
- **Install:** OK to install now. Safe to add mid-save.

7.4 Installation Checkpoint

 OK to install all mods in this chapter now. Safe to add mid-save.

7.5 Next Steps

Proceed to [Chapter 8: Gameplay Enhancements](#).

8 Chapter 8: Gameplay Enhancements

8.1 Learning Objectives

- Reduce tedious micromanagement
- Improve pawn AI to make smarter decisions automatically
- Add quality-of-life features that deepen strategy without making the game easier

8.2 Overview

8.3 The Mods

8.3.1 Smarter Construction

- **What it does:** Prevents pawns from building themselves into corners or blocking incomplete structures. Pawns will avoid walling themselves in, and they'll build from the outside in so they aren't trapped.
- **Why it's here:** Anyone who's played RimWorld has had a pawn finish a wall segment and realise they're trapped inside a room with no door. Smarter Construction prevents this by having pawns build in a logical order — open edges first, enclosed spaces last. Great for filling sapper tunnels and rebuilding breached walls.
- **Not OP because:** Only changes build order logic — doesn't affect build speed, material costs, or skill requirements. Right-click → Construct intentionally bypasses the mod so you can still trap raiders.
- **Install:** OK to install now. Safe to add/remove mid-save. Fully compatible with Blueprints, Misc. Robots++, and Achtung!.

8.3.2 Keep Converting

- **What it does:** Set a target for your Moral Guide to auto-cast Convert whenever it's off cooldown until the target converts or becomes invalid. Configurable check interval (default every 2500 ticks).
- **Why it's here:** Converting prisoners in vanilla requires you to manually monitor cooldowns and re-cast Convert every time it's ready. Miss a cycle and the target regains certainty. Keep Converting automates this tedious cycle so you can focus on running your colony.
- **Not OP because:** The Convert ability still has its full cooldown and base success chance — nothing is sped up or made more effective. The mod just removes the need to manually re-cast every time. If the target dies, leaves the map, or converts, the auto-cast stops.
- **Install:** OK to install now. Safe to add/remove mid-save.

8.3.3 Pharmacist: Represcribed

- **What it does:** Lets you configure medicine usage rules per injury severity — set which medicine quality (herbal, industrial, glitterworld) doctors can use for minor cuts, major wounds, diseases, and ongoing conditions. A continuation/fork of Fluffy's original Pharmacist mod.
- **Why it's here:** Vanilla doctors use the best available medicine for every treatment — they'll crack open glitterworld medicine for a scratch. Pharmacist lets you reserve high-quality medicine for serious injuries and surgery while using herbal medicine for minor wounds. Saves your rare resources without micromanaging every bandage.
- **Not OP because:** Medicine consumption still happens at the same rate — you're just using the right tier for the right job. The mod doesn't create medicine, reduce consumption, or improve treatment quality. It's logistics, not power.
- **Install:** OK to install now. Safe to add mid-save. Doctors will respect the configured thresholds for medicine choice.

8.3.4 Self Dyeing

- **What it does:** Enables an option at the Styling Station that lets pawns autonomously dye their clothes over time. Configurable preferences — favour ideology colour, favourite colour, or a mix. Supports custom dye patterns.
- **Why it's here:** Manually dyeing every piece of apparel for every colonist is tedious and easy to neglect. Self Dyeing lets pawns handle their own wardrobe over time, gradually shifting clothes to your preferred colour scheme without any player input after the initial setup.
- **Not OP because:** Dyeing still consumes dye resources from your stockpile. Pawns dye clothes gradually (not instantly). The mod only automates an existing manual process — no free resources, no stat changes.
- **Install:** OK to install now. Safe to add mid-save. Requires a Styling Station to be built and configured.

8.3.5 Fluffy Breakdowns (1.6 Update)

- **What it does:** Adds a maintenance system — powered buildings (refrigerators, comms consoles, fabricators, etc.) gradually accumulate breakdowns and need periodic repairs. A new “Maintenance” bill type lets you schedule fixing tasks. Breakdown severity escalates if ignored.
- **Why it's here:** Vanilla buildings never break down, which means once you build something, you never think about it again. Breakdowns create a reason to maintain your colony — giving your constructors steady work between projects and adding a layer of logistics to mid-game base management.
- **Not OP because:** Breakdowns progress slowly and give plenty of warning. The maintenance work is mild — it doesn't distract from survival. It adds depth without punishing the player. By Fluffy (reputable mod author known for balanced, well-designed mods).
- **Install:** OK to install now. Buildings will start accumulating wear. Safe to add mid-save.

8.3.6 Eccentric Tech - Fusion Power

- **What it does:** Adds the Multi-Mode Fusion Stellarator (MMFS) — a compact fusion reactor with multiple fuel modes (deuterium, tritium, helium-3), fuel extraction/production buildings, and ignition capacitors. 4 new research projects, 5 new resources, 7 new buildings.
- **Why it's here:** Mid-to-late game power demands grow fast — especially with modded turrets, shields, and production chains. Fusion Power gives you a scalable, realistic upgrade path beyond vanilla geothermal/solar, without resorting to cheat-level generators.
- **Not OP because:** Reactors require costly startup (ignition capacitors), consume industrial fuels that must be produced or traded, and hull integrity degrades over time — forcing maintenance shutdowns. Pushing a degraded reactor risks a plasma vent that sets everything on fire.
- **Install:** OK to install if you have a stable mid-game colony with fabrication unlocked. Best on a new save due to world-gen trade ships.

8.3.7 Hacking Expansion

- **What it does:** Adds a hacking system with a craftable cyberdeck — hack mechanoid clusters, enemy turrets, and ancient structures to unlock superior technologies. Introduces new threats and challenges alongside the rewards. No DLC required, but integrates with all of them.
- **Why it's here:** Vanilla hacking is limited to a few niche interactions. This mod turns hacking into a full gameplay system with meaningful risk-reward decisions: hack a cluster for rare tech? It might alert more mechanoids. The progression from basic cyberdeck to master hacker gives you a new vector for mid-to-late-game power advancement.
- **Not OP because:** Hacking requires crafting the cyberdeck first, then researching upgrades. Failed hacks have consequences — alerting enemies, damaging the cyberdeck, or triggering defensive countermeasures. The unlocked tech is powerful but gated behind risk. You can't hack everything immediately.
- **Install:** OK to install now. Safe to add mid-save.

8.3.8 Mech Chargers Repair Mechs

- **What it does:** Vanilla mech chargers (Biotech DLC) slowly repair mechanoids while they recharge — healing injuries and restoring destroyed parts over time. Configurable repair speed (default 10 HP per in-game hour).
- **Why it's here:** In vanilla, repairing damaged mechanoids requires manual intervention — crafting repair paks, ordering mechs to specific repair stations, and babysitting the process. This mod makes mech chargers pull double duty: charge and repair in one step. The repair is slow enough that it won't save a mech mid-battle, but it keeps your mechanoid army ready between engagements without tedious micro.
- **Not OP because:** Damaged mechs don't automatically path to chargers — you still need to manually order them to recharge. Repair speed is configurable (1–50 HP/hour) and happens while the mech is charging (meaning it's not available for work or combat). Restoring destroyed parts can be toggled off in settings if it feels too generous. Requires Biotech DLC.
- **Install:** OK to install now. Safe to add mid-save. Requires Biotech DLC. Works with both regular and large vanilla mech chargers.

8.4 Installation Checkpoint

 OK to install all mods in this chapter now.

8.5 Next Steps

Proceed to [Chapter 9: Furniture & Production](#).

9 Chapter 9: Furniture & Production

9.1 Learning Objectives

- Expand your base-building options with vanilla-friendly furniture sets
- Add production depth with specialised workstations and linkables
- Customise rooms with medical, security, power, and decorative furniture

9.2 Overview

RimWorld's vanilla furniture is functional but limited. The Vanilla Furniture Expanded series adds carefully balanced furniture modules that integrate seamlessly — new production buildings, structural options, defensive emplacements, medical equipment, power generators, spacer-tier workbenches, and decorative items. Each module is independent; install only the ones that suit your playstyle.

All mods in this chapter require the Vanilla Expanded Framework (installed in Chapter 3).

9.3 The Mods

9.3.1 Vanilla Furniture Expanded - Production

- **What it does:** Adds electric versions of manual workbenches (stone cutting, butchering, drug production), 6 new linkable buildings that provide work-speed boosts when placed next to matching workstations, and deep drills, quarries, and industrial-scale production buildings.
- **Why it's here:** Vanilla production peaks once you build tool cabinets everywhere. Linkables create a reason to organise workstations into dedicated manufacturing rooms instead of a single workspace blob. The electric butcher table and stone cutter are direct upgrades that save pawn walking time.
- **Not OP because:** Linkable bonuses stack with tool cabinets but require adjacency planning. Electric versions cost components and research. Quarries produce stone slowly and deplete over time.
- **Install:** OK to install now. Safe to add mid-save. Requires Vanilla Expanded Framework.

9.3.2 Vanilla Furniture Expanded - Architect

- **What it does:** Adds new walls (reinforced, animated, transparent), fences, gates, bridge types (stone, metal, wooden), foundations for building on sand, cattle gates, mooring posts, support pillars, and expanded floor and carpet colours. All items are vanilla-fitting — no RGB sliders, just hand-picked colours that match the game's palette.
- **Why it's here:** Vanilla structures are limited to one wall type, a few floors, and no bridges over deep water. Architect gives you functional variety — build over water, reinforce vulnerable walls, create livestock pens with cattle gates, and design docks with mooring posts for trade ships. The support pillars let you create open-plan interiors with realistic structural support.
- **Not OP because:** All structures cost the same or slightly more than their vanilla equivalents. Reinforced walls take more damage but cost more materials. Bridges and foundations have appropriate costs and cannot be built everywhere.
- **Install:** OK to install now. Safe to add mid-save. Requires Vanilla Expanded Framework.

9.3.3 Vanilla Furniture Expanded - Power

- **What it does:** Adds upgraded power generation options — industrial solar panels, high-yield wind turbines, chemfuel generators with fuel refining, upgraded geothermal vents, new battery types, and power switches with logic control. Each upgrade is a separate research project.
- **Why it's here:** Mid-game power demands outgrow vanilla solar and wind quickly, and geothermal is limited by vent placement. VFE Power gives you a progression: refine chemfuel for dedicated generators, build denser solar arrays, and eventually stabilise your grid with industrial batteries. The power switch logic lets you prioritise critical systems during spikes.

- **Not OP because:** Every generator upgrade costs more resources and research. Industrial solar panels produce more power but require more space and components. Chemfuel generators need a steady fuel supply chain. Batteries still degrade and can short-circuit.
- **Install:** OK to install now. Safe to add mid-save. Requires Vanilla Expanded Framework. Complements Eccentric Tech - Fusion Power (Chapter 8) — VFE Power covers the mid-game, Fusion covers the late-game.

9.3.4 Vanilla Furniture Expanded - Security

- **What it does:** A full defensive overhaul adding embrasures, trenches, dugouts, razor wire, bear traps, ballistas, catapults, manned machine-gun nests, flamer turrets, tesla coils, railguns, and more — spanning medieval through spacer tech levels. Rebalances and replaces the original security module.
- **Why it's here:** Vanilla defensive options are sparse (sandbags and vanilla turrets). VFE Security gives you meaningful choices at every tech tier: medieval walls and traps before electricity, trench warfare at industrial, and energy weapons at spacer. The manned turret complexes let your gunners contribute directly rather than relying solely on automatics.
- **Not OP because:** Every defensive structure costs resources and research. Manned turrets require a pawn operating them — they can't fight and work simultaneously. Trenches reduce mobility. Embrasures provide cover but limit line of sight. Raid strength scales with colony wealth, so better defenses invite stronger raids.
- **Install:** OK to install now. Safe to add mid-save. Requires Vanilla Expanded Framework. Works alongside Reel's Turret Pipeline (Chapter 10) — VFE Security provides the turrets, Turret Pipeline handles ammo logistics.

9.3.5 Vanilla Furniture Expanded - Medical Module

- **What it does:** Adds hospital beds, vital monitors, IV stands, medicine cabinets, surgery lights, wheelchairs, sterile walls, and colour-coded floor markings. New linkable buildings boost surgery success and treatment speed when placed near medical beds.
- **Why it's here:** Vanilla medical furniture is a single bed type and a vitals monitor. This module gives you a proper hospital buildout — IV stands for long-term patient care, medicine cabinets for organised storage, surgery lights for operating rooms, and wheelchairs for mobility-impaired pawns. The sterile walls and colour markings let you design a professional infirmary.
- **Not OP because:** Linkable bonuses require proper room layout. Wheelchairs consume resources to craft. All medical buildings cost components and take space. Surgery lights and monitors only improve outcomes when paired with skilled doctors — they don't replace training.
- **Install:** OK to install now. Safe to add mid-save. Requires Vanilla Expanded Framework.

9.3.6 Vanilla Furniture Expanded - Spacer Module

- **What it does:** Adds spacer-tier furniture and production buildings — advanced research benches, sterile fabrication benches, plasteel storage units, hi-tech shelves, spacer beds, and specialised workstations that require spacer tech research to build.
- **Why it's here:** Once you reach spacer tech, your base still uses the same furniture as day one. The Spacer Module gives your high-tech colony a visual and functional upgrade — sterile, efficient workstations that match your technological progression. The advanced storage and research benches directly support late-game production chains.
- **Not OP because:** All buildings require spacer tech research and advanced materials (plasteel, components, sometimes uranium). They're sidegrades and efficiency improvements, not leaps — your pawns still need the same skills to use them.
- **Install:** OK to install now for established mid-game colonies. Requires Vanilla Expanded Framework.

9.3.7 Vanilla Furniture Expanded - Props and Decor

- **What it does:** Adds decorative items — rugs, curtains, banners, flower pots, wall art, decorative pillars, fencing, tapestries, and dozens of static props for beautifying rooms without functional gameplay impact.
- **Why it's here:** Vanilla decoration options are limited to sculptures, flowers, and a few art pieces. Props and Decor gives you the tools to make rooms feel lived-in — hang banners in your throne room, place curtains in bedrooms, roll out rugs in the dining hall, and line your walkways with decorative fencing. Beauty values are balanced against vanilla sculptures.
- **Not OP because:** Purely cosmetic. Decorative items have beauty values comparable to vanilla equivalents of the same material and quality. They take up space, cost materials, and serve no functional purpose beyond room aesthetics.
- **Install:** OK to install now. Safe to add mid-save. Requires Vanilla Expanded Framework.

9.4 Installation Checkpoint

 OK to install all mods in this chapter now. They are safe to add mid-save. All require the Vanilla Expanded Framework (already installed from Chapter 3).

9.5 Next Steps

Proceed to [Chapter 10: Weapons & Combat](#).

10 Chapter 10: Weapons & Combat

10.1 Learning Objectives

- Expand the arsenal with balanced new weapons
- Improve combat AI and tactics
- Keep combat challenging — not easier

10.2 Overview

10.3 The Mods

10.3.1 Defensive Positions

- **What it does:** Lets colonists remember their designated defensive positions. Save up to 4 positions per colonist (Advanced mode) and send everyone to their posts with one click. Also provides hotkeys for quick group selection and positioning.
- **Why it's here:** In a raid, every second counts. Vanilla requires you to manually draft each pawn and click them to their position — time you don't have when sappers are breaching your wall. Defensive Positions lets you pre-plan positions during downtime and deploy instantly when the alarm sounds.
- **Not OP because:** It doesn't make pawns fight better — it just gets them to the right place faster. The positions are static (pawns won't re-position mid-battle), and you still need to manage tactics once they're there. Requires HugsLib.
- **Install:** OK to install now. Safe to add mid-save. Requires HugsLib (already in your load order from Chapter 3).

10.3.2 Eccentric Tech - Advanced Shields

- **What it does:** Adds 3 advanced shield belts (Phased, Modulated, Layered) with unique mechanics — increased capacity, ranged-fire-through, and multi-layer shatter barriers. Crafted at the Nanoassembler.
- **Why it's here:** Vanilla shield belts are one-size-fits-all. These give you tactical choices: tank more damage but can't recharge mid-fight (Phased), shoot while shielded (Modulated), or shrug off heavy hits but die to fast fire (Layered).
- **Not OP because:** Each belt has a meaningful downside — longer recharge delay, lower total capacity, or vulnerability to certain weapon types. They require Eccentric Tech - Core and significant research to craft. CE-compatible natively.
- **Install:** OK to install now. Safe mid-save, but requires Eccentric Tech - Core.

10.3.3 Eccentric Extras - Flares

- **What it does:** Adds 10 illumination tools (torches, flares, flare guns, chemlights) and 7 utility apparel items (flare packs, chemlight packs, leg-pack variants). Also adds illumination mortar shells.
- **Why it's here:** Night battles and indoor fighting are brutally punishing without light. These give you tactical options to light up dark areas — affecting accuracy, work speed, and surgery success — without needing to build powered lamps everywhere.
- **Not OP because:** Flares and torches are consumable. Chemlights last longer but produce dimmer light. Auto-use logic has quirks (pawns may waste flares near doors). They illuminate rather than deal damage — purely tactical utility.
- **Install:** OK to install now. Safe to add mid-save.

10.3.4 Pulse Weaponry

- **What it does:** Adds 3 spacer-level pulse weapons (rifle, SMG, minigun), a pulse helmet, and a pulse turret. All share a distinct energy-weapon aesthetic and projectile effect.

- **Why it's here:** Late-game combat is dominated by charge rifles and lances. Pulse Weaponry diversifies the spacer-tier arsenal with balanced alternatives that have distinct fire patterns and trade-offs.
- **Not OP because:** Weapons are balanced against existing charge/plasma counterparts. The mod is focused and small — no power creep, just variety. Requires Athena Framework for projectile effects.
- **Install:** OK to install now for mid-to-late game colonies. Weapons require spacer tech research.

10.3.5 Spacer Arsenal

- **What it does:** Adds 7 new spacer-level guns with varied roles (Photon SMG, Thermite Shotgun, Clash HMG, etc.), plus a marine helmet and armor set.
- **Why it's here:** Gives the spacer tech tier more depth and variety — different weapons for different combat roles instead of “charge rifle for everyone.” Integrates into the existing Pulse-charged munitions research.
- **Not OP because:** Each weapon has its own niche with clear strengths and weaknesses. They're sidegrades to vanilla charge weapons, not upgrades. No required dependencies.
- **Install:** OK to install now for mid-to-late game colonies. Integrates with existing research.

10.3.6 Reel's Turret Pipeline

- **What it does:** Adds an ammunition pipe network that automatically refills connected turrets. Build ammo pipes from your storage to defensive positions — turrets pull ammo directly from the network instead of needing a pawn to manually rearm them.
- **Why it's here:** Manually rearming turrets is tedious and dangerous — pawns get shot lugging shells across the killbox. The pipe system turns turret maintenance from a constant chore into a one-time setup, letting you focus on tactics.
- **Not OP because:** You still need to produce the ammo. Pipes cost resources to build and space to route. A severed pipe line stops feeding until repaired. It automates logistics, not combat — your turrets still fire the same ammo at the same rate.
- **Install:** OK to install now. Safe to add mid-save.

10.4 Installation Checkpoint

10.5 Next Steps

Proceed to [Chapter 11: World Variety](#).

11 Chapter 11: World Variety

11.1 Learning Objectives

- Add new biomes, factions, and events to make every playthrough unique
- Expand exploration and world interaction
- Increase replayability without overpowering the player

11.2 Overview

11.3 The Mods

11.3.1 More Vanilla Biomes

- **What it does:** Adds 10 vanilla-friendly biomes including Glacial Shield, Badlands, Cloud Forest, Marsh, Stormy Savanna, Coastal Dunes, Sandbar, Alpine Meadow, Oasis, and Ice Floes. Each has distinct terrain, flora, and wildlife distributions.
- **Why it's here:** RimWorld's vanilla biomes are limited and can get repetitive. More Vanilla Biomes slots seamlessly into world generation with biomes that feel like they belong — no sci-fi gimmicks, just more variety for starting location choices.
- **Not OP because:** New biomes have balanced terrain and resource distributions. Some are harder (Ice Floes), some easier (Alpine Meadow). They don't add op items or mechanics. Has built-in compatibility patches for many animal and plant mods.
- **Install:** OK to install now, but best experience on a new world save. Requires Harmony. Compatible with most biome and world-gen mods.

11.3.2 Real Ruins

- **What it does:** Adds ruins of real player-built bases to your world maps. These aren't premade templates — they're actual bases uploaded by other players on a shared server. Sometimes they contain loot, sometimes threats, sometimes nothing at all. Also includes world-quest events for exploring full abandoned bases.
- **Why it's here:** Vanilla ruins are repetitive — the same few prefab layouts every time. Real Ruins turns every map into a potential discovery. Finding another player's failed freezer room or half-built killbox tells a story without a single line of dialogue. The exploration depth adds hours of map-by-map variety.
- **Not OP because:** Ruin quality and contents are weighted by your colony's wealth. Early-game ruins contain little of value. The server distributes blueprints impartially — no curated "good" ruins, just whatever real players uploaded. You always need to weigh risk (bugs, traps, turrets) against reward.
- **Install:** OK to install now. Best on a new world save (ruins generate during map creation). Requires internet access for the ruin server. Safe to add mid-save (new maps will generate ruins, existing maps won't).

11.3.3 Sometimes Raids Go Wrong

- **What it does:** Adds random twists to raids — meteorites strike the raiders, manhunting packs were tracking them, mechanites cause rapid ageing, and other chaotic events. Each twist triggers instead of (or during) a raid, changing the encounter unpredictably.
- **Why it's here:** After a few raids, they all blur together — same factions, same approach, same outcome. Sometimes Raids Go Wrong turns every raid into a unique story moment. A pack of manhunting boomrats chasing your raiders into your killbox creates a memory no scripted event can match. The mod has been actively maintained since 2017, with continuous 1.6 updates.
- **Not OP because:** The twists affect raiders unpredictably — they can help you (meteorites killing raiders) or hurt you (enraged raiders with new abilities). Nothing is guaranteed. The events are

controlled by storyteller weights, so the frequency is configurable. You can't control which twist fires or when.

- **Install:** OK to install now. Safe to add mid-save. Check mod settings to adjust event weights if certain twists feel too frequent.

11.3.4 Leo - Events

- **What it does:** Adds 60+ new events including 30+ conditional (temporary) environment changes — biome expansions, dancing plagues, migrating forests, awakened trees, silver showers, unexpected mineral seams, washed-up flotsam, and many more. Each event changes your colony's situation for a set duration.
- **Why it's here:** Vanilla event variety is thin — you cycle through the same dozen incidents every playthrough. Leo - Events more than doubles that pool with creative, often bizarre scenarios that force you to adapt. A migrating forest creeping across your map or a dancing plague incapacitating half your colony creates emergent stories that keep the mid-game fresh. The mod explicitly prioritises gameplay over pure immersion, meaning events are designed to be fun and interactive rather than just atmospheric.
- **Not OP because:** Events are temporary and balanced through the storyteller's threat cycle. Good events (silver showers, mineral seams) give resources but often attract attention or require work to harvest. Bad events (dancing plague, awakened trees) are disruptive but survivable with proper preparation. Every event has a clear end condition.
- **Install:** OK to install now. Safe to add mid-save. With 60+ events, consider reviewing the event list in mod settings to disable ones that don't fit your playstyle.

11.4 Installation Checkpoint

✅ OK to install all mods in this chapter now. Real Ruins generates on new maps; Sometimes Raids Go Wrong and Leo - Events trigger dynamically on existing saves.

11.5 Next Steps

Proceed to [Chapter 12: Monsters & Creatures](#).

12 Chapter 12: Monsters & Creatures

12.1 Learning Objectives

- Add new animal species, monsters, and threats to the world
- Balance new creatures so they're challenging but fair
- Integrate with existing RimWorld mechanics (taming, hunting, combat)

12.2 Overview

12.3 The Mods

12.3.1 Vanilla Animals Expanded

- **What it does:** Adds dozens of vanilla-friendly animals — from house cats and rottweilers to anacondas and Indian elephants — spread across every biome. This is the consolidated pack that replaces all the old individual biome modules. Every animal can be individually disabled in mod settings.
- **Why it's here:** RimWorld's base animal pool is small and every biome feels the same — the same muffalo, the same rats, the same bears. Vanilla Animals Expanded is the foundation of creature variety. Each biome gets unique fauna: tropical rainforests have anacondas and gorillas, arid shrublands have ostriches and warthogs, cold biomes have yaks and arctic foxes. Animals are designed to match vanilla art and balance precisely, so they slot in as if Ludeon added them.
- **Not OP because:** Every animal is balanced against vanilla equivalents — stats, hunger rate, tameness, and combat ability are all within vanilla ranges. The author maintains a public spreadsheet of every animal's exact stats. You can disable any animal you don't want. Requires Vanilla Expanded Framework (already in your load order).
- **Install:** OK to install now. Safe to add mid-save. Animals will begin spawning on new maps and via events. Use mod settings to disable any animals you don't want.

12.3.2 Alpha Animals

- **What it does:** Adds over 100 unique creatures with distinct mechanics — walking resource factories, indestructible plant monsters, darkness-dwelling stalkers, teleporting ambushers, and a new insectoid faction called the Black Hive. Each creature introduces something new: special leathers, unique organs, unusual behaviours.
- **Why it's here:** Where Vanilla Animals Expanded adds familiar wildlife, Alpha Animals adds the truly alien. Its creatures are designed around the philosophy that every animal should bring a new gameplay mechanic. The Black Hive faction diversifies raid types. Creatures spawn per-biome or appear only through traders, encouraging exploration. It's the most popular creature mod in the RimWorld community for good reason.
- **Not OP because:** Powerful creatures are proportionally rare. The Gallatross (a walking resource generator) is extremely hard to tame and requires advanced handling skill. The Black Hive's unique drops are gated behind surviving their raids. Every creature's spawn rate and stats are configurable in mod settings. Requires Harmony and Vanilla Expanded Framework (both already in your load order).
- **Install:** OK to install now. Safe to add mid-save — new creatures will populate newly-generated maps. Consider adding the Alpha Animals Retextured mod for improved 512×512 textures.

12.3.3 Megafauna

- **What it does:** Adds 38 prehistoric species — woolly mammoths, smilodons, titanoboas, paraceratheriums, and more — that spawn naturally in appropriate biomes. Animals are rare, challenging to tame or hunt, and offer valuable resources (wool, milk, eggs, unique leathers). Many can serve as pack animals.

- **Why it's here:** Prehistoric creatures fill a gameplay gap that vanilla doesn't cover: megafauna that are genuinely threatening to mid-game colonies. A pack of deinotheriums or a territorial titanoboa creates memorable encounters that scale with colony progression. Mammoth wool and sivatherium milk give caravan-focused colonies a reason to invest in taming. The mod is exceptionally well-maintained — updated for every RimWorld version since Beta 18 (2017) with active 1.6 support.
- **Not OP because:** Megafauna spawns are intentionally rare. Large herbivores yield lots of meat and leather but are hard to take down and even harder to tame. Predators are dangerous but avoidable with proper scouting. All spawn rates and commonality are configurable in mod settings. No dependencies required.
- **Install:** OK to install now. Safe to add mid-save. Animals will spawn on newly-generated maps. If you use A Dog Said... or Giddy-Up!, Megafauna auto-detects and patches them.

12.3.4 Vanilla Animals Expanded — Waste Animals

- **What it does:** Adds biome-specific waste creatures for polluted environments — Toxbears, Wastefallos, Pestigators, Megatardis, and more. Unlike vanilla Biotech (which only spawns Toxalopes and Toxrats everywhere), this mod tailors waste animals to their habitat. Also adds a new event: generate too many wastepacks and waste animals swarm your map.
- **Why it's here:** If you play with Biotech DLC (which this guide assumes), polluted biomes are currently monotonous — the same two creatures everywhere. Waste Animals makes pollution a dynamic gameplay factor: different biomes have different toxic creatures, and your waste management directly affects what shows up. The swarm event turns pollution from a passive storage problem into an active threat, reinforcing the DLC's core loop.
- **Not OP because:** Waste animals are biome-specific and tied to pollution levels — clean colonies rarely see them. The swarm event scales with your waste production, so you control the trigger. Creature stats are balanced against vanilla Biotech animals. Requires Vanilla Expanded Framework and Biotech DLC.
- **Install:** OK to install now. Best on a new world save to ensure proper biome distribution. Safe to add mid-save for new maps.

12.3.5 A Dog Said... Animal Prosthetics 2

- **What it does:** Adds prosthetics and bionics for animals — peg legs, simple prosthetic limbs, full bionics, and organ replacements. Animals are categorized by size/intelligence to determine which parts they can receive (rats get peg legs, elephants get full bionics). Also allows healing permanent animal injuries with medicine.
- **Why it's here:** With 100+ new creatures from the mods above, your animals will eventually get hurt. Vanilla's animal medicine is limited to basic tending — scars and missing body parts are permanent. A Dog Said... 2 lets you treat your animal companions the same way you treat your colonists. The research and material costs mean it's a mid-game investment, not a day-one upgrade. This is the continuation of the iconic original mod, actively maintained for 1.6.
- **Not OP because:** Animal bionics require dedicated research (gated behind human prosthetics), crafting at an Animal Prosthetics Table, and significant resources. The category system prevents small animals from receiving end-game bionics — you can't turn a rat into a cyber-warrior. Scar healing requires industrial medicine. Built-in patches cover all Vanilla Animals Expanded creatures.
- **Install:** OK to install now. Safe to add mid-save. Requires XML Extensions (already in your load order) for the optional category override system. Compatible with all animal mods listed in this chapter. Not compatible with the original A Dog Said... (use this version only).

12.4 Installation Checkpoint

✅ OK to install all mods in this chapter now. They are additive — new creatures will spawn on newly-generated maps and via events on existing saves. Safe to add mid-save, but for the best distribution of biome-specific animals, start a new world.

12.5 Next Steps

Proceed to [Chapter 13: NPC Interactions](#).

13 Chapter 13: NPC Interactions

13.1 Learning Objectives

- Increase depth of faction relationships and diplomacy
- Expand quest variety and storytelling possibilities
- Make the world feel more alive with NPC interactions

13.2 Overview

13.3 The Mods

13.3.1 More Creepjoiners

- **What it does:** Adds 8+ new creepjoiner types to the Anomaly DLC. Includes the Tree Devourer (eat special trees for stacking buffs), Void Touched (craft indestructible voidmetal walls, transmute bioferrite limbs), and Highlander (steal abilities from other creepjoiners, killing them in the process). Each has unique abilities, drawbacks, and research projects.
- **Why it's here:** Vanilla creepjoiners get repetitive fast — you see the same few types every playthrough. More Creepjoiners makes Anomaly encounters feel fresh by introducing genuinely different joiners with real trade-offs. The Void Touched's voidmetal walls are a meaningful defensive tool without being a win button because they require Anomaly research first.
- **Not OP because:** Every creepjoiner has clear downsides. The Tree Devourer sacrifices trees your colony might need. The Highlander must kill other creepjoiners to steal their powers (losing their benefits permanently). Voidmetal walls require significant research investment and bioferrite. Requires Anomaly DLC.
- **Install:** OK to install now. Safe to add mid-save.

13.4 Installation Checkpoint

13.5 Next Steps

Proceed to [Chapter 14: Alien Breeding](#).

14 Chapter 14: Alien Breeding

14.1 Learning Objectives

- Understand RimWorld's genetic engineering systems
- Genetically modify colonists and animals
- Breed alien hybrids with unique traits
- Customize xenotypes with new genes
- Add new xenohuman races to your world

14.2 Overview

RimWorld's Biotech DLC introduced genetic engineering, letting you extract, combine, and implant genes into colonists. The mods in this chapter take that foundation and run with it — you'll splice animal DNA into hybrid monsters, mutagen your colonists into furry (or scaly) hybrids, selectively breed super-animals, and add a dozen new xenohuman races to the rim.

This is mid-to-late-game territory. Genetic manipulation requires research benches, specialized buildings, rare resources, and time. The payoff is real: custom-engineered colonists, hybrid war-beasts, and a colony that's truly yours. None of these mods are OP because they're gated behind research, resources, and risk.

14.3 Section 1: Genetics Framework

These four mods form the mechanical backbone of alien breeding. Vanilla Genetics Expanded is the hybrid-crafting core; Alpha Genes floods the gene pool with variety; Pawnmorpher adds mutagenic mayhem; Animal Genetics gives you a reason to care about which muffalo breeds with which.

14.3.1 Vanilla Genetics Expanded

- **What it does:** Adds a complete genetic engineering system — extract genomes from any creature, splice them into genoframes, and gestate hybrid animals in growth vats. Create everything from a bear with a boomrat's explosive death to a thrumbo that lays eggs. Includes new research projects, gene banks, genome sequencers, and Abandoned Genetics Lab quests to explore. Also adds a hostile hybrid faction that raids you with their own creations.
- **Why it's here:** This is the definitive RimWorld genetics mod — the successor to the original Genetic Rim, rebuilt from the ground up by the same author (Sarg Bjornson). It gives you a proper laboratory workflow: capture creatures, extract their genome, design a hybrid, grow it. The Abandoned Lab quests turn gene-hunting into exploration. The hostile hybrid faction means your own tools can be turned against you, creating a satisfying arms-race dynamic.
- **Not OP because:** Every hybrid requires a genoframe (costs research and rare resources to craft), gestation in a growth vat (takes days), and a compatible gene donor. Powerful combinations require high-quality genoframes made from multiple extractions. The hostile hybrid faction scales with your tech level — they show up with nastier hybrids as you advance. Requires Vanilla Expanded Framework (already in your load order).
- **Install:** OK to install now. Safe to add mid-save — the VE Framework will prompt you to add the new hybrid faction when you load.

14.3.2 Alpha Genes

- **What it does:** Adds over 300 new genes to the Biotech gene system, 16 new xenotypes with unique mechanics, and a Custom Random xenotype system that lets you design your own xenotype and have it spawn naturally in custom factions. Also adds Abandoned Biotech Lab quests (separate from VGE's) and genetic tools that let you tweak colonists' genes mid-game.
- **Why it's here:** Vanilla Biotech includes about 80 genes. Alpha Genes multiplies that by four — you get new metabolic traits, cosmetic genes, combat augments, production specializations, and downright weird ones (mineral eaters, photosynthesis, toxic blood). The 16 new xenotypes (like

the plant-like Florans, the stone-skinned Lithoids, and the psionic Starchildren) add world-building depth. The Custom Random system is the sleeper hit: design a xenotype, assign it to a faction, and watch it spread naturally.

- **Not OP because:** Genes are balanced within the Biotech economy — powerful genes have high metabolic cost, negative trade-offs, or both. The Lab quests require you to clear dangerous sites. Random xenotype generation respects complexity limits. No new genes are strictly better than vanilla equivalents — they're sidegrades and specializations. Requires Biotech DLC and Vanilla Expanded Framework.
- **Install:** OK to install now. Safe to add mid-save. New genes will appear in genepacks from traders, quests, and world generation. Also consider Alpha Genes Integrated (3538704272) which blends Alpha Genes into vanilla xenotypes and genepack distribution.

14.3.3 Pawnmorpher

- **What it does:** Adds a full mutagen system — craft serums that give pawns animalistic mutations (horns, wings, tails, muzzles, fur), turn people into hybrid animal-people, or mutate entire raids into confused livestock. Includes mutagenic mortar shells, IED traps, and rifles. Mutations develop over time, starting weak and growing stronger. Also adds stabilizers and reverters for risk management, plus new events (mutagenic ship chunks, mutonite meteors, cowflu).
- **Why it's here:** Pawnmorpher is the mad-scientist mod. Where VGE is about precision gene-splicing, Pawnmorpher is about chaotic mutation — you don't know exactly what you'll get, and sometimes your best researcher ends up as a cat. The mutagen mortar is a genuine tactical option: fire a green cloud into a raid and watch raiders turn into harmless livestock. The sapience system means transformed pawns can retain their personality, letting you roleplay a colony of talking animals.
- **Not OP because:** Mutations are unpredictable — you can't target specific traits, and every mutation carries the risk of full animal transformation. Stabilizers are expensive. Reverters have a limited window before mutations become permanent. The mortar shells require mutagen production and have a cooldown. Requires Humanoid Alien Races 2.0 and HugsLib (both available on the Workshop).
- **Install:** OK to install now. Safe to add mid-save. Pawnmorpher: VAE (2875883329) adds morphs for Vanilla Animals Expanded creatures if you want even more variety.

14.3.4 Animal Genetics (Continued)

- **What it does:** Adds seven inheritable genetic traits to animals — Health, Damage, Speed, Capacity (carrying), Meat yield, Leather yield, and Milk/Wool production. Offspring inherit values from their parents with a mutation factor. Adds a Genetics tab showing every animal's stats and a wildlife overview. Configurable to also affect human reproduction.
- **Why it's here:** Vanilla RimWorld has no inheritance system — every animal is born identical to its parents. Animal Genetics turns animal husbandry into a real mechanic: selectively breed muffalos for higher carrying capacity, train fighting bears with bonus damage, or create a super-cow that produces double milk. The mutation system means you can't just line up two perfect parents and get perfect offspring — there's always variation, which keeps breeding interesting. Compatible with all animal mods (Alpha Animals, Megafauna, Vanilla Animals Expanded).
- **Not OP because:** Gene values can be above or below 100% — you might breed stronger animals but also accidentally breed weaker ones. The mutation factor is configurable. Selective breeding takes many generations and requires managing herds. Human inheritance is optional and off by default. Requires only Harmony (already in your load order).
- **Install:** OK to install now. Safe to add mid-save. Consider The Birds and the Bees or Neuter for population control alongside selective breeding.

14.4 Section 2: Xenotype Races (Vanilla Races Expanded)

Vanilla Races Expanded is a series of mods that enhance Biotech's xenotype system. Each adds a new xenohuman race with unique genes, mechanics, items, and world presence. They all require Biotech DLC and Vanilla Expanded Framework (both already covered).

You don't need to install all of them. Pick the ones that fit your story. They're all safe to add mid-save, but you should never remove a xenotype mod from an active save.

14.4.1 Vanilla Races Expanded — Sanguophages

- **What it does:** Expands sanguophages into three sub-xenotypes — Strigoi (classic vampires), Ekkimian (shadow-dwellers), and Bruxa (feral blood-hunters). Adds blood types and affinities (O, A, B, AB with matching bonuses), Blood Moon events that empower vampires but attract threats, the Hemohunter mental state, new deathrest building enhancements, and a full hemogen economy system.
- **Why it's here:** Vanilla sanguophages are fun but shallow — one xenotype, one build. VRE Sanguophage turns vampirism into a colony theme. The blood type system means you care about which pawns feed whom. Blood Moons create memorable set-piece events. The three sub-xenotypes let you choose your vampire fantasy: elegant immortals, skulking shadows, or feral hunters.
- **Not OP because:** Every sanguophage still requires hemogen management and deathrest. Blood types create dependencies — if your only blood donor leaves or dies, your vampires starve. Blood Moons escalate over time. Bruxae are powerful in combat but harder to control. All the original sanguophage weaknesses (sunlight sensitivity, deathrest) remain.
- **Install:** OK to install now. Safe to add mid-save. New sanguophages will generate with the expanded system; existing ones are unaffected.

14.4.2 Vanilla Races Expanded — Android

- **What it does:** Adds androids as a full xenotype — mechanical pawns that don't eat, sleep, or breathe. They require maintenance instead of food, recharge instead of sleeping, and can be repaired rather than healed. Includes new maintenance stations, charging stations, android-specific implants (processing cores, sensory arrays), and an Android faction.
- **Why it's here:** If you've ever wanted a colony that never sleeps, never starves, and never has mental breaks, androids are your answer. They fill a niche that neither baseliners nor mechanoids cover: pawns that work 24/7 without complaints. The maintenance system replaces the food/sleep economy with a resource economy (steel, components), which creates different gameplay decisions than biological colonists.
- **Not OP because:** Androids can't gain passions or skills naturally — they start with fixed stats based on their model. Maintenance costs steel and components, which are finite early-game. They can't reproduce — you build them or buy them. They take damage differently and can't use conventional medical treatment for everything. A colony of pure androids misses out on the storytelling that comes from biological pawns' needs and relationships.
- **Install:** OK to install now. Safe to add mid-save. New android factions will appear in world generation.

14.4.3 Vanilla Races Expanded — Genie

- **What it does:** Overhauls the Genie xenotype with new genes that emphasize their role as fragile geniuses. Genies are brilliant at crafting and intellectual work but have no natural immunity — they require antibiotics to survive. Adds antibiotics crafting, new genie-specific prosthetics (precision manipulators, neural accelerators), and an expanded lore system.
- **Why it's here:** Vanilla Genies are boring — they're just "good at crafting, bad at fighting." VRE Genie gives them a real gameplay identity: high-risk, high-reward specialists who can outperform any baseliner at skilled work but require constant medical support. The antibiotics system adds a

new production chain and a reason to maintain a pharmaceutical industry. They're the glass cannons of the xenotype world.

- **Not OP because:** No natural immunity means even a scratch can become fatal without antibiotics. Antibiotics require research and pharmaceutical production. Genies are physically fragile — they go down faster in combat and from diseases. The mod explicitly warns it makes the game harder. A genie colony is a constant medical management challenge.
- **Install:** OK to install now. Safe to add mid-save. This mod increases difficulty — consider waiting until you're comfortable with baseline genies.

14.4.4 Vanilla Races Expanded — Hussar

- **What it does:** Expands Hussars with new combat-focused genes and adds a new xenotype — Uhlans, luciferium-fueled super-soldiers. Uhlans are permanently dependent on luciferium but gain extreme resilience, speed, and combat power. Adds new genes (Adrenaline Rush, Pain Nullification, Combat Reflexes), new weapons, and a dedicated Hussar faction.
- **Why it's here:** Hussars are RimWorld's canonical supersoldiers, but vanilla they're one-note. VRE Hussar gives them depth: the Uhlan variant is a genuine Faustian bargain — incredible power in exchange for permanent luciferium addiction. The new faction gives you Hussar-exclusive traders and quests. The combat genes can be extracted and applied to other pawns, making Hussars a genetic resource worth capturing alive.
- **Not OP because:** Uhlans require luciferium — one of the rarest and most expensive resources in the game. If your supply chain breaks, Uhlans suffer catastrophic withdrawal. Hussar/Uhlan factions are hostile to most other factions, limiting trade options. The powerful combat genes come with metabolic costs. Requires Biotech DLC and Vanilla Expanded Framework.
- **Install:** OK to install now. Safe to add mid-save. The new faction will be added via VE Framework prompt.

14.4.5 Vanilla Races Expanded — Waster

- **What it does:** Overhauls Wasters with a mutation system — Wasters gain random beneficial mutations when exposed to pollution, creating unique pawns over time. Adds toxic weaponry, waste-based crafting, and a Waster faction with their own settlements. Mutations are permanent and stack, so a long-lived Waster becomes genuinely alien.
- **Why it's here:** Vanilla Wasters are pollution-tolerant but boring. VRE Waster makes pollution a resource rather than a problem — settle next to a toxic waste dump and your Wasters will evolve unique traits. The mutation system ties directly into the chapter's breeding theme: Wasters are the xenotype that changes over time, shaped by their environment. If you play with pollution-heavy mods or Biotech wastepacks, Wasters become genuinely interesting.
- **Not OP because:** Mutations are random — you can't control what your Wasters develop. Some mutations are negative. Wasters still require pollution tolerance management. The Waster faction is hostile to clean-environment factions. Polluted bases have health risks for non-Waster colonists.
- **Install:** OK to install now. Safe to add mid-save. Best on a save that generates pollution (Biotech wastepacks, polluted maps).

14.4.6 Vanilla Races Expanded — Fungoid

- **What it does:** Adds Fungoids — an archite-dependent xenotype that gradually loses mental capacity while gaining physical power. Over time, Fungoids become stronger, tougher, and faster but lose skills, social ability, and eventually sapience. Includes a Ship Part Crash event where a fungoid ship crashes on your map, new genes (Mind Coalescence, Gene Infector, Fungal Crest), fungal weapons, and a full lore system.
- **Why it's here:** Fungoids are the most unique VRE xenotype — they're not a static race but a degenerative transformation. A Fungoid colonist is a ticking clock: powerful now, but deteriorating mentally. This creates compelling storytelling: do you keep your favorite colonist as

a Fungoid for the combat power, or cure them and lose the benefits? The infection mechanic means Fungoids can spread to other colonists, creating an outbreak-management subgame.

- **Not OP because:** Mental degeneration is inevitable and permanent — high-level Fungoids can't do skilled work, socialize, or follow complex orders. The transformation is difficult to reverse. The Ship Part Crash event is a significant threat. Requires Vanilla Expanded Framework and Biotech DLC.
- **Install:** OK to install now. Safe to add mid-save. New fungal events will start appearing after loading.

14.4.7 Vanilla Races Expanded — Phytokin

- **What it does:** Adds Phytokins — plant-human hybrids that photosynthesize for food and reproduce by planting a saplingchild. Three varieties based on DLC: Poluxkin (base, evil-aligned), Gauranlenkins (with Ideology, tree-connection themed), and Animakins (with Royalty, psychic-plant themed). Phytokins have extreme psychic sensitivity and affinity with plant life.
- **Why it's here:** Phytokins are the most thematically distinct VRE xenotype — they're not humans with tweaks, they're a different form of life. Their sapling reproduction system is genuinely alien: instead of pregnancy, a Phytokin plants a child that grows over days. Their photosynthesis means they don't need conventional food, making them ideal for barren biomes. The three DLC-based variants mean the mod scales with your DLC collection.
- **Not OP because:** Phytokins are extremely rare — you'll need to complete a quest to find your first one. They have extreme psychic sensitivity (bad for psychic drone events). Sapling reproduction takes time and leaves the sapling vulnerable. They're fragile in melee. Photosynthesis requires light — they struggle in darkness, caves, and toxic fallout. Requires Biotech DLC and Vanilla Expanded Framework.
- **Install:** OK to install now. Safe to add mid-save. A quest will appear to rescue your first Phytokin after adding the mod.

14.4.8 Vanilla Races Expanded — Starjack

- **What it does:** Expands the Starjack xenotype (added by Odyssey DLC) into a fully space-adapted species. Adds the Astrogene system — genes that only activate in space, offering powerful but situational bonuses. New genes include Vacuum-Impervious, Cryostasis, Vacuum Efficiency, and Spacejump Tail. Starjacks now have Oxygen Sensitivity traits that make planetary life increasingly hazardous, reinforcing their identity as voidborn specialists.
- **Why it's here:** The Odyssey DLC's Starjacks are interesting but underdeveloped — they're adapted for space but don't feel like it mechanically. VRE Starjack fixes that: on a ship or space station, Starjacks are unparalleled. On the ground, they struggle. This creates a compelling narrative of a xenotype that's perfectly adapted to a life your colonists probably don't have — at least until you build a starship. The randomized Astrogene system means every Starjack is unique.
- **Not OP because:** Astrogens only work in space — on a planetary map they're inactive. Oxygen Sensitivity makes planetary life progressively harder. Starjacks struggle in atmospheric environments. Requires Odyssey DLC, Biotech DLC, and Vanilla Expanded Framework.
- **Install:** OK to install now. Safe to add mid-save. Only install if you have the Odyssey DLC — the mod does nothing without it.

14.5 Installation Checkpoint

All mods in this chapter are safe to add mid-save, but consider these guidelines:

- **Vanilla Genetics Expanded, Alpha Genes, Pawnmorpher, Animal Genetics** — OK to install any time. New genetics content will appear via research and quests.
- **Vanilla Races Expanded mods** — Safe to add mid-save. Each adds its xenotype to world generation. Genepacks containing their genes will appear from traders.

- **Never remove a VRE mod mid-save.** Xenotype mods embed data in your save file that can't be safely cleaned.
- **Start a new game** if you want the full experience — new factions, xenotype distribution, and world-gen content will be properly seeded.

14.6 Next Steps

Proceed to [Chapter 15: Gameplay Guide](#).

15 Chapter 15: Gameplay Guide — A Full Modded Run

15.1 Learning Objectives

- Apply all previous mod chapters in a cohesive playthrough
- Understand how mods interact and compound over a full game
- Navigate early, mid, and late game with 100+ mods

15.2 Overview

15.3 Starting Your Modded Colony

15.4 Early Game (Days 1-30) — With Mods

15.5 Mid Game (Days 30-120) — With Mods

15.6 Late Game (Days 120+) — With Mods

15.7 Common Modded Pitfalls

15.8 Next Steps

Proceed to [Chapter 16: Troubleshooting](#).

16 Chapter 16: Troubleshooting

16.1 Learning Objectives

- Diagnose common mod-related issues
- Fix load order problems
- Recover from crashes without losing your colony
- Optimize performance when the game slows down

16.2 Overview

16.3 Common Issues

16.3.1 Game Crashes on Startup

16.3.2 Red Errors in Dev Mode

16.3.3 Performance Issues

16.3.4 Mods Not Working

16.3.5 Save File Issues

16.4 Getting Help

16.5 Next Steps

Congratulations! You've completed the guide. See [Appendix C: Resources](#) for community links and further reading.

17 Appendix A: Complete Modlist Reference

#	Mod Name	Chapter	Author	Steam Workshop ID
1	Harmony	03 — Framework	pardeike	2009463077
2	HugsLib	03 — Framework	UnlimitedHugs	818773962
3	XML Extensions	03 — Framework	UnlimitedHugs	2577315200
4	Vanilla Expanded Framework	03 — Framework	Oskar Potocki	2023507013
5	Eccentric Tech - Core	03 — Framework	Aelanna	2552623545
6	Camera+	05 — UI / HUD	Brrainz	867467808
7	BetterInfoCard	05 — UI / HUD	Skullywag	2890920739
8	Damage Indicators [1.6]	05 — UI / HUD	Jaxxa	2016331497
9	Allow Tool	05 — UI / HUD	UnlimitedHugs	761421485
10	RimHUD	05 — UI / HUD	Jaxe	1508850027
11	Better Workbench Management	05 — UI / HUD	fluffy	935982361
12	Designator Shapes	05 — UI / HUD	Merthykins	1235181370
13	Dynamic Trade Interface	05 — UI / HUD	Soki	3020706506
14	Range Finder	05 — UI / HUD	Brrainz	1332119637
15	Power Tab 2	05 — UI / HUD	Owlchemist (fixed)	3528086894
16	DDI - Deep Drill Indicator	05 — UI / HUD	Georodin	2566736688
17	RandomPlus	05 — UI / HUD	mastertea	1434137894
18	Day Counter	05 — UI / HUD	AlenDio	3490487928
19	Fahrenheit and Celsius	05 — UI / HUD	Xeo	937759575
20	Tainted Renaming	05 — UI / HUD	Divinorium	2021251001
21	CCP's Stone Tweaks Vanilla	06 — Graphics	Canute	1524681306
22	Floor-Mounted Aquarium	06 — Graphics	Mlie	3538928291
23	AI Upscaled Textures - Core	06 — Graphics	Cosmo Corban	2307494117

#	Mod Name	Chapter	Author	Steam Workshop ID
24	Vanilla Pawns Retextured	06 — Graphics	Neronix17	2275310562
25	Expedition 33 Music	07 — Audio & Music	Warult	3498135045
26	No Time Change Sounds	07 — Audio & Music	rhadamant	2572368517
27	Smarter Construction	08 — Gameplay	Daz	2202185773
28	Keep Converting	08 — Gameplay	Linnun	3461478214
29	Pharmacist: Represcribed	08 — Gameplay	Ilysen	3527418098
30	Self Dyeing	08 — Gameplay	avil	2562859859
31	Fluffy Breakdowns (1.6 Update)	08 — Gameplay	Fluffy	2584064444
32	Eccentric Tech - Fusion Power	08 — Gameplay	Aelanna	2742125879
33	Hacking Expansion	08 — Gameplay	Unknown	3573344880
34	Vanilla Furniture Expanded - Production	09 — Furniture & Production	Oskar Potocki	1880253632
35	Vanilla Furniture Expanded - Architect	09 — Furniture & Production	Oskar Potocki	2608762624
36	Vanilla Furniture Expanded - Power	09 — Furniture & Production	Oskar Potocki	2062943477
37	Vanilla Furniture Expanded - Security	09 — Furniture & Production	Oskar Potocki	1845154007
38	Vanilla Furniture Expanded - Medical Module	09 — Furniture & Production	Oskar Potocki	1718191613
39	Vanilla Furniture Expanded - Spacer Module	09 — Furniture & Production	Oskar Potocki	2028381079
40	Vanilla Furniture Expanded - Props and Decor	09 — Furniture & Production	Oskar Potocki	2102143149
41	Defensive Positions	10 — Weapons/ Combat	UnlimitedHugs	761219125

#	Mod Name	Chapter	Author	Steam Workshop ID
42	Eccentric Tech - Advanced Shields	10 — Weapons/ Combat	Aelanna	2591721581
43	Eccentric Extras - Flares	10 — Weapons/ Combat	Aelanna	2552628275
44	Pulse Weaponry	10 — Weapons/ Combat	DetVisor	3077879089
45	Spacer Arsenal	10 — Weapons/ Combat	DetVisor	3247891820
46	Reel's Turret Pipeline	10 — Weapons/ Combat	Reel	3424132769
47	More Vanilla Biomes	11 — World Variety	Zylleon	1931453053
48	More Creepjoiners	13 — NPC Interactions	Unknown	3434682604
49	Real Ruins	11 — World Variety	mrorangu	1552146295
50	Sometimes Raids Go Wrong	11 — World Variety	RemingtonRyder	1551336515
51	Leo - Events	11 — World Variety	Leo39994	3730168769
52	Forced Rearmament	04 — Bugfixes & Performance	seeki	3232549170
53	Mech Chargers Repair Mechs	08 — Gameplay	NotherKiller	3721671407
54	Vanilla Animals Expanded	12 — Monsters & Creatures	Oskar Potocki	2871933948
55	Alpha Animals	12 — Monsters & Creatures	Sarg Bjornson	1541721856
56	Megafauna	12 — Monsters & Creatures	Spino	1055485938
57	Vanilla Animals Expanded — Waste Animals	12 — Monsters & Creatures	Oskar Potocki	2962126499
58	A Dog Said... Animal Prosthetics 2	12 — Monsters & Creatures	Samuel_Bucher	3238353862
59	Animal Variety Coats	06 — Graphics	cucumpear	1511926373

18 Appendix B: Advanced & Optional Mods

The mods in this appendix introduce major new gameplay systems that add significant complexity. They are not recommended for a first playthrough — come back to them once you're comfortable with the core modlist.

18.1 Vanilla Vehicles Expanded

18.1.1 Vanilla Vehicles Expanded

- **What it does:** Adds drivable vehicles — cars, trucks, APCs, tanks, and aircraft. Vehicles can be built, fuelled, repaired, and upgraded. They provide fast travel on the world map, mobile bases, vehicle combat, and cargo transport. Each vehicle type has unique stats, weapon mounts, and upgrade slots.
- **Why you'd want it later:** Once you've mastered RimWorld's core loop, vehicles add a whole new strategic layer. Fast caravan travel, mobile forward operating bases, armoured hit-and-run tactics, and airborne raids change how you interact with the world. The vehicle crafting and fuel chain adds depth to mid-late game production.
- **Not OP because:** Vehicles cost significant resources to build and maintain (fuel, components, repairs). They require dedicated pawns with crafting skills to maintain. Vehicle combat is risky — a damaged vehicle can explode, killing the crew. Raid strength still scales with wealth, including vehicle value.
- **Install:** Wait until you're comfortable with the base modlist. Requires **Vehicle Framework** (3014915404), Vanilla Expanded Framework, and Harmony. Best on a fresh save due to world-gen integration.

18.1.2 Vanilla Vehicles Expanded - Upgrades

- **What it does:** Adds upgrade modules for vehicles — armour plating, weapon mounts, engine tunings, cargo expanders, sensor arrays, and specialised equipment. Vehicles can be customised for different roles (scout, transport, assault) using the upgrade system.
- **Why it's here:** Vanilla Vehicles Expanded gives you the vehicles; Upgrades lets you tailor them to your needs. A scout jeep needs speed and sensors; a troop transport needs armour and weapon mounts. The upgrade system extends the vehicle tech tree and gives your crafters more to do.
- **Not OP because:** Upgrades cost resources and require research. Heavier upgrades slow the vehicle. Each vehicle has limited upgrade slots, forcing trade-offs. Over-upgrading can reduce reliability and increase fuel consumption.
- **Install:** Only after installing Vanilla Vehicles Expanded. Safe to add mid-save.

18.2 Other Notable Advanced Mods

18.2.1 Vanilla Factions Expanded - Mechanoids

- **What it does:** Adds a full mechanoid faction with unique base types, raid patterns, and endgame threats. Mechanoid bases generate on the world map and escalate in difficulty as your colony grows.
- **Why you'd want it later:** Once you're comfortable with vanilla raids, the mechanoid faction adds a distinct enemy type with its own AI behaviour and tactical requirements. Fighting mechanoids requires different strategies than human raiders — EMP weapons, shield belts, and careful positioning become essential.
- **Not OP because:** The faction scales with colony wealth and threat level. Mechanoid bases can be avoided or scouted before engaging. Provides additional challenge without being unfair — requires the same tactical preparation as any late-game threat.
- **Install:** Best on a fresh save for world-gen integration. Requires Vanilla Expanded Framework.

18.3 Next Steps

Ready for more? Return to [Chapter 10: Weapons & Combat](#) or continue to [Chapter 11: World Variety](#).

19 Appendix C: Resources

19.1 Official Resources

- [RimWorld Official Wiki](#)
- [RimWorld Forums](#)
- [RimWorld Discord](#)

19.2 Modding Tools

- [RimPy Mod Manager](#)
- [Steam Workshop - RimWorld](#)
- [RimWorld Mods on GitHub](#)

19.3 Learning Content

- [r/RimWorld](#) — community, showcases, guides
- [Nookrium YouTube](#) — RimWorld mod showcases
- [ambiguousamphibian YouTube](#) — entertaining playthroughs
- [Francis John YouTube](#) — strategic deep dives
- [Pete Complete YouTube](#) — excellent playthroughs

19.4 Tools Used in This Project

- [mdbook](#) — builds this guide
- [Pandoc](#) — PDF rendering engine